BEVY_ZEROVERSE: AUDITABLE PROCEDURAL INDOOR DATA FOR MULTI-VIEW RECONSTRUCTION

A PREPRINT

Mitchell Mosure
mitchell@mosure.me

October 2026

ABSTRACT

We describe an open-source synthetic data engine combining primitive-object scenes with furnished procedural interiors, calibrated multi-view cameras, optional human motion, and aligned geometric annotations. Indoor generation samples continuous architectural, furnishing, appearance, lighting and camera parameters under geometric constraints. A continuous camera-baseline control coordinates spacing, shared-surface overlap and path variation. The architectural evaluation audits 512 consecutive rooms and 256 aligned views across 32 rooms. A matched camera study covers 64 rooms, 1–12 cameras and 3 baseline values, with native shared-surface measurements from 2 prespecified room families. Exact configurations, capture-engine identities and reproducible artifacts accompany the measured cohorts. Photographic realism, unlimited-process stability and downstream pretraining gains are not established by these diagnostics.

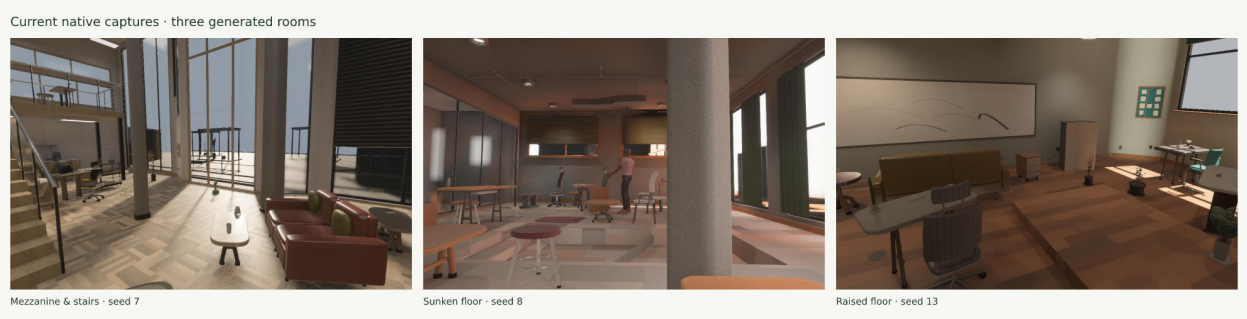


Figure 1: Current generated architecture: mezzanine stairs, a sunken lounge and a raised platform with an arched portal. Native RGB at $t = 0$; seeds 7, 8 and 6, cameras 3, 3 and 1. Plans, calibrated views, aligned annotations and the unfiltered cohort follow in the main evaluation.

1 Purpose and scope

Sparse-view reconstruction benefits from observations of shared surfaces: cameras that see unrelated objects offer little direct correspondence supervision. LRM-Zero demonstrates the potential of procedurally synthesized object data [1]; MVSplat uses cross-view matching to estimate geometry [2]. Our contribution is an inspectable data-generation system and its bounded evaluation, rather than a trained reconstruction model or a reconstruction accuracy result.

The engine supports Object, CornellCube, Room, SemanticRoom and Human scene modes alongside furnished interiors in `ProceduralIndoor`. The runtime uses Bevy 0.20, optional Burn 0.22 inference, and published AnnyBody/ARDY integrations. Office, conference, lounge, classroom, library and related activity priors influence a compositional program; they do not select complete meshes or a finite collection of room templates. Most architecture, furniture,

plants and clutter require neither mesh repositories nor photographic texture collections. AnnyBody human meshes and optional motion/embedding models are explicit pretrained dependencies.

2 Procedural scene program

A scene is generated as $S = G(s, \theta)$, where s seeds independent random streams and θ contains user-controlled densities, activity, quality, camera and motion policies. The serialized manifest records the sampled program, object transforms and seeds, people, photometry and camera paths. A separate generator version and capture-engine identity invalidate incompatible resumptions. Determinism means reproducing the same program and settings with compatible assets and engine versions; bit-identical rasterization across GPU vendors is not promised.

2.1 Architecture, objects and appearance

Room area and aspect, functional zones, aisle widths and activity mixtures are sampled continuously. The exterior footprint supports taper, chamfered corners, a concave cut-in and up to two offset returns on eligible side edges. Accepted returns preserve the complete shared wall to the neighboring room and at least 78% of the original footprint area. A shared architectural program controls polygon triangulation, wall apertures, sloping ceilings, closed soffits, arched portals, pillars, floor levels and furnished mezzanines with stairs and guards. Furniture support, camera clearance, fixture anchors and voxel export use the same surfaces, including local headroom beneath soffits. Windows vary aperture dimensions, sill/head heights, bay widths, frames, inset depth and shades across multiple exterior edges. An internal glazed connection retains a furnished neighboring room. This is a bounded architectural grammar: footprints are connected simple polygons without interior holes, neighboring rooms and mezzanine decks remain rectangular, and arbitrary multi-storey connectivity, structural analysis and building-code compliance are outside scope.

Furniture is assembled from material-specific parts. Tables vary outline, taper, thickness and leg/pedestal construction; chairs vary seat shell, curvature, back proportions, recline, arms, base and headrest. Sectional seating, shelving, potted plants, outlets, switches, ceiling lights, whiteboards, displays and tabletop objects extend the semantic inventory. Placement uses support relationships, clearance checks, zone-dependent furnishing fields and bounded rejection. Chairs sample partial and deep insertion beneath tables, including occupied seats; the rendered leg, pedestal, brace and panel program also supplies collision geometry. Working poses mix free reaches with one- or two-palm tabletop support using fixed-length arm IK and skin-aware contact fitting. This is static surface support, not simulated grasping or animated contact dynamics. Surface clutter varies shared arrangement, handedness, spacing and local jitter. Object yaw and arrangement are sampled rather than restricted to orthogonal grids. Collision proxies approximate detailed meshes; geometric validity is checked separately from physical plausibility.

Surfaces use 36 role-specific base material slots with procedural albedo, normal and roughness maps, metric UV scales and mip chains. Recipes vary spatial frequency, directional grain, textile structure, mineral veining, relief, wear, color and roughness. Upholstery finish slots share three independently seeded structure programs per substrate; clothing reuses these maps with independent actor pigmentation and atlas scaling. Woven textiles sample over/under grammars, yarn diameter, slub, crimp, dyed warp/weft and fuzz. Leather varies grain spacing, stretch, pore relief, nap, pigmentation and polish. Coupled albedo, metric normal, cavity and roughness fields are baked into mipmapped textures. Textile directional GGX scattering approximates aggregate fibre reflection; it is not a measured cloth multiple-scattering BSDF. Mineral programs blend cement binder, exposed aggregate, pinholes and warped multiscale vein deposits with independent polish. Wall finishes vary roller stipple, sprayed and flattened plaster relief, trowel marks and gloss; glazed ceramics add restrained crackle and a protective lobe. Timber separates grain cuts from stain absorption, bleaching, pore fill and varnish; face and edge grain share their finish. Foliage atlases keep veins aligned to the blade fold and vary chlorophyll patches, pale variegation, wax and thin-surface transmission. Timber and foliage finish slots share two texture structures per role; metallic, chrome and plastic finishes use three continuously varied structure programs. Paint, neutral knit and the other supported substrates bake their maps on the GPU once per scene, retaining CPU recipe oracles and coupled normal/roughness mipmaps. Timber and stone floor atlases use 512-square maps to resolve separately cut boards and stone tiles; other substrates use 256-square maps. Screens, boards and printed pages have independently seeded content. Only referenced finish variants are built during runtime preparation. Glass recipes vary roughness, refractive index, tint, attenuation and thickness. Direct lights vary placement, intensity, color temperature, circuit activity and direction; exposure, daylight and sky radiance are independently recorded. Low-light samples are retained in evaluation.

Timber samples offset and tapered growth-ring cuts, uneven earlywood/latewood, elongated pores and sparse branch knots; floor planks have independent cuts. Unresolved normal variation contributes to filtered GGX roughness. Chrome retains a conductive base layer with continuously varied polished/brushed roughness and anisotropy; micron scratches are represented as scattering rather than quantized visible bumps. A static scene-derived HDR environment traces the

rendered triangle assemblies and visibility-tested direct lights using the shared transport BVH. Linear radiance is stored in float16 cubemaps, with GGX specular and cosine diffuse convolution; source footprints suppress sparse-emitter filtering noise. Secondary bounce energy is approximate. Native rendering uses at most three box-projected zone probes and a global fallback; WebGPU uses the global environment because localized probes require texture-array capabilities. These bounded probes improve spatial reflection placement but do not provide live moving-human reflections or full specular path tracing. Their cosine/GGX convolutions execute once on the GPU with a CPU oracle; the static radiance tracing remains on the CPU.

People retain AnnyBody geometry with continuous phenotype/pose parameters, fitted garments, surface hair and facial details. Static people remain the default. Opt-in ARDY motion combines seeded procedural text sequences with object-aware waypoint planning and scene-time playback. Requesting static scenes does not initialize motion inference. Motion models use the upstream loaders and caches; cached prompt text does not replace the motion seed. Clothing, hair, joint limits and collision handling are approximations, particularly during complex motion.

2.2 Multi-view sampling and training defaults

Camera centers and complete trajectories are constrained to the primary furnished room by default. The sampler varies focal length, position, target, roll and path shape, checks obstacle clearance and prefers visible content. Static capture uses one timestep; optional trajectories support repeated calibrated observations. A reference camera anchors a connected set of views. For each other camera and five trajectory times, we require

$$\min\{o_{0\to i}(t), o_{i\to 0}(t)\} \geq \tau, \qquad b_{\min} \leq \|c_i(t) - c_0(t)\| \leq b_{\max},$$

where o estimates first-surface overlap using visibility rays and object/architecture proxies. Overlap constrains reference edges. An additional minimum separation applies to every camera pair at 33 synchronized path samples. Group spread is the horizontal camera-center covariance's minor/major standard-deviation ratio. Moving pairs must differ in normalized RMS displacement after removing their starting offsets. Full-path swept collision checks remain; sampled spread and overlap are not continuous-time or rendered-pixel guarantees.

The camera-baseline control $b \in [0, 1]$ expands into explicit advanced settings: minimum pair separation $0.04 + 0.46b$ metres, minimum reference distance $0.06 + 2.30b$ metres, maximum reference distance $0.35 + 9.65b$ metres, overlap target $0.65 - 0.50b^{0.7}$, group spread $0.20 + 0.10b$, independent path variation b, and visible-region target variation b. The default is $b = 0.5$. Wide settings permit distant viewpoints and lower overlap; narrow settings favor close views of shared content. Collision and visibility rejection affect the realized distribution. Arbitrary room, camera-count and constraint combinations need not be feasible.

Peers can aim at different first-surface proxy hits visible to the reference camera. Target interpolation and proposal frequency increase with baseline; frequency also increases with camera count. Feasible candidates are ranked for nearest-peer distance, optical-axis separation and additional quantized primary-room surface cells. Preferred distance scales with room diagonal divided by the square root of camera count; it is not a hard packing constraint. Selection considers at most 2, 3 or 4 feasible candidates for groups of 2–4, 5–8 or 9+ views, within a 12-proposal window after the first feasible candidate and the existing total search budget. Zero target variation retains shared-target, first-fit rigs. Rare completion for 2–11 views can use one bounded overcomplete search, retaining the reference and rechecking every subset against the same geometric constraints. All selected views still satisfy the original collision, content, distance and overlap gates. Proxy surface-cell novelty does not measure exact mesh area or reconstruction gains. Related temporal routes and occasional shared-target recovery remain supported; varied gaze does not imply independent route topology.

The inspector exposes this slider above the advanced controls. Any advanced edit switches to explicit custom values. Changes take effect on regeneration, so dragging a control does not rebuild the room. JSON input accepts `{"baseline":0.75}` within the `indoor_camera` configuration. Serialization records the expanded values, avoiding hidden preset precedence. Camera travel length and long-route frequency are separate from spacing between cameras. Joint collision and overlap rejection can favor shorter accepted routes at wide spacing; the minimum travel bound remains explicit. The finite search budget reports impossible settings without silently weakening them.

3 Capture and annotation contract

Capture begins only after scene preparation, assets, shader pipelines, lighting and requested human trajectories are ready. A bounded asynchronous readback protocol associates outputs with the requested scene, camera, time and mode; it does not infer readiness from an arbitrary number of frames. Sequential native indoor streams automatically prepare future rooms on a shared bounded CPU pool and can upload one future room's immutable assets during current readback, preserving the RGB settling boundary. Speculative material and indirect-lighting compute waits until every current

Indoor training configuration	Default and rationale
Capture cameras / timesteps	Four / one; synchronized views of static geometry
Shared-view policy	$b = 0.5$: $\tau \approx 0.342$, pair separation ≥ 0.27 m, reference distance 1.21–5.175 m; explicit `multiview:null` enables independent views
Group diversity	Horizontal spread ratio ≥ 0.25 for 3+ views; moving-pair normalized RMS difference ≥ 0.075
Room/program/density	Primary room; Mixed samples all 20 programs uniformly; furnishing density 0.65, human density 0.25
Camera path policy	0.03–8 m allowed path length, 0.45 long-path proposal fraction; collision rejection still applies
Human motion	Disabled; static people remain, no ARDY initialization
Native lighting	Auto quality, shadows and approximate baked diffuse GI, 256 rays/probe
Dataset CLI workers / chunks	One process, four rooms/chunk; explicit overrides retained
Finite worker lifetime	256 scenes/child for finite indoor jobs with process workers
Default export	RGB and calibration; geometric channels are opt-in

Table 1: Defaults conditional on the procedural-indoor dataset CLI scene. Other scene modes retain their defaults. These are an engineering starting point, not an optimum established by downstream training. Library callers explicitly choose camera and worker counts.

capture copy is submitted. Future entities remain excluded from the current capture. Between ordinary live requests, one declared successor can complete preparation with capture cameras disabled; this consumes the existing lookahead budget. This scheduling retains full native quality rather than reducing shadows or geometry. Geometric channels share an RGBA32Float ground-truth raster attachment. RGB uses Bevy's PBR pipeline, shadow maps, tone mapping and optional indirect-lighting approximation. Float32 storage does not imply that all RGB or lighting computations have float32 intermediate precision: HDR color and irradiance volumes include float16 stages.

Capture-camera supersampling defaults to an automatic policy: images below 1280×720 pixels use a $2\times$ linear raster scale, while larger images use native resolution. Explicit off, $2\times$, $3\times$ and $4\times$ overrides are available. A GPU box resolve averages RGB in linear light after tone mapping into the requested output dimensions, without a CPU filtering round trip. Numeric depth, normals, position, semantics, flow and co-visibility retain the exact final pixel-center rays. They are not averaged across surfaces: RGB can have mixed edge coverage while its geometric target describes the central ray. Calibration and flow units describe the final grid; versioned sampling metadata records the resolved factor. Filtered annotation previews are display products, not replacements for these geometric targets. Supersampling increases raster work and GPU attachment memory without increasing final capture readback bytes.

Headless generation supports chunked safetensors or per-sample files, compression, resumable identity checks, metrics and bounded worker replacement. RGB and numeric channels have separate encoding contracts. Native generation performs numeric temporal export and O-Voxel export. The WebGPU viewer supports indoor construction, inspection and optional motion inference, using the same manifest and geometry modules; Portable quality reduces unsupported or costly rendering features. Browser velocity visualization is not a qualification of native dataset flow export. The current Wasm path omits native baked diffuse GI and uses its platform lighting fallback.

3.1 Co-visibility representation and surface agreement

Co-visibility shares the float32 surface raster used by the other geometric annotations. For source camera s, pixel p and time t, the exported unsigned 16-bit integer is

$$M_s(p,t) = \sum_{\substack{j=0 \\ j \neq s}}^{N-1} 2^j V_{s \to j}(p,t), \qquad 1 \leq N \leq 16,$$

where $V_{s \to j} \in \{0,1\}$ records visibility of the source surface in peer camera j. The source camera's own bit is always zero; the editor has no membership bit. The ordered camera legend remains identical across views and timesteps in a sample. A peer's binary supervision is recovered by testing $(M_s \,\&\, 2^j) \neq 0$; the number of observing peers is $\mathrm{popcount}(M_s)$. Thus the integer preserves camera identities, unlike a shared-pixel count alone.

The GPU pass projects a valid source surface into each peer, checks near/far clipping and image bounds, and reads the nearest target pixel. For source point x, target point y, corresponding world-space unit normals n_s, n_j and target axial depth d, the surface-agreement test is

$$\max(|(y-x)\cdot n_s|, |(y-x)\cdot n_j|) \leq \max(0.001\,\mathrm{m}, 10^{-4}d).$$

Output	Meaning and qualification
Color	Display RGB; lossless float storage or explicitly chosen codecs. Separate linear-HDR export supports renderer comparisons.
Depth	Colorized, normalized or metric linear depth; axial/ray conventions; calibration records the selection.
Normal / position	View-space normals encoded as $(n+1)/2$; world positions affine-normalized by the exported AABB, without clamping. First surface, float32 geometric path.
Semantic	Shared class vocabulary, aligned with depth/normal/position, including ceiling light fixtures.
Optical flow	Forward displacement $(\Delta u, \Delta v)$ in pixels, with correspondence-valid and target-visible masks. Terminal time has no forward correspondence.
Motion vectors	The same exported displacement divided by image width/height. Viewer velocity colors are previews, not these numeric annotations.
Co-visibility	Optional same-time, per-pixel membership in other capture cameras. A uint16 mask assigns one bit per camera (up to 16), with a separate source-valid mask and a consistent additive RGB legend; the source camera and editor are excluded.
Calibration / instances	Camera intrinsics, poses and normalized trajectory time, scene manifest, semantic instance oriented boxes, person joints, and primary-room annotation AABB.
O-Voxel	Primary-room architecture/objects only; excludes neighboring/outside geometry. Requires one timestep and disabled human motion; incompatible requests are rejected.

Table 2: Eight currently supported render modes plus structured annotations. The architectural cohort exports six modes, including production co-visibility; optical flow and motion vectors are available for temporal dataset export but are not measured in this cohort. Glass is a first geometric surface by default. The opt-in through-glass policy excludes transmissive hits consistently from geometric, semantic and co-visibility annotations along unrefracted rays; RGB transmission is unchanged. Flow transports surface correspondence, including supported human deformation, rather than estimating photometric image motion.

This local-plane test avoids comparing depths along different pixel-center rays on sloped surfaces. It remains resolution- and tolerance-dependent: silhouettes, thin geometry, curved surfaces and near-contact surfaces can be ambiguous. Pixel-level membership need not be symmetric after raster reprojection. The chosen solid or through-glass annotation policy applies to the source and every peer. Reflected and refracted RGB content does not imply co-visible geometry.

The additive RGB8 preview sums the contributions of set camera bits without tone mapping. For four cameras, the shared legend is $C_0 = (170, 0, 0)$, $C_1 = (0, 255, 0)$, $C_2 = (0, 0, 255)$ and $C_3 = (85, 0, 0)$. For example, $M = 10 = \texttt{0b1010}$ means peers 1 and 3, with color $(85, 255, 0)$. These colors encode camera membership, not semantic classes or overlap magnitude. The palette adapts deterministically to camera count and remains uniquely decodable for all supported combinations in the original lossless PNG.

A separate uint8 source-valid mask A_s distinguishes an unshared surface ($A_s = 1, M_s = 0$) from background ($A_s = 0, M_s = 0$); both appear black in the additive preview. Folder exports retain membership and validity as $[H, W, 1]$ arrays in NPZ. Chunked exports use $[B, T, N, H, W, 1]$. Training should consume these numeric arrays and the camera metadata, rather than rasterizing the PDF or decoding a resized or lossy preview. Shared percentages divide by valid source-pixel observations, including unshared surfaces, not unique 3D points.

4 Procedural pipeline and measured cost

Figure 2 follows the production path from seeded programs to completed views. Immutable architecture, furnishing and person assemblies are constructed once. The same triangles supply transport and rendering. One shared pool with at most eight dedicated preparation threads executes geometry, materials, tangents and both transport trees; scoped callers can also execute queued work. Ordered triangle emission releases the assembly borrow before tree construction and mesh realization proceed concurrently, without cloning complete scenes. Ordered results and reserved mesh handles preserve seed identity. Material preparation overlaps geometry transport, group planning and mesh/tangent realization. Only referenced material variants are prepared. Supported material maps and their mip chains are baked once on the GPU; remaining programs use the CPU oracle. Native room lookahead overlaps preparation, bounded speculative uploads and static lighting with current capture, without promoting future scene entities early. The browser uses cooperative yields in place of native worker parallelism; yields return control between stages but do not provide CPU parallelism or guarantee a frame-time budget.

4.1 Protocol and accounting

Measurements use Intel Core i9-12900K (24 logical CPUs), NVIDIA RTX PRO 6000 Blackwell Workstation (96 GiB VRAM), Linux Vulkan; release subprocesses pinned to logical CPUs 0 and 2. Native development profile with opt-level 3 and debug information disabled, nightly rustc 1.100; frozen source-matched executables; automatic SSAA and full native quality. Active production training shared the CPU and GPU with explicit operator approval. One release renderer ran at a time with two-CPU affinity, a 10 GiB task RSS cap and free-memory floors. Two short capture repetitions are a shared-workstation screen, not isolated hardware throughput or long-run memory qualification. These are shared-machine measurements, not isolated hardware limits or a training-throughput benchmark. The receipt binds source SHA256 `3c21395218bb18c4`, package versions, exact configuration, input hashes and every measured seed. No geometry, texture resolution, shadow quality or GI-ray budget is reduced for this study. Static AnnyBody people are sampled at density 0.25; ARDY inference is disabled. Auto SSAA renders the three 512×512 RGB outputs at 1024×1024 before the GPU box resolve; numeric annotations retain the 512×512 pixel-center grid.

The CPU-only study samples 32 consecutive rooms, seeds 400–431, with 2 separate warm-up rooms. It invokes the same preparation entry point as capture, including camera sampling, layout validation, geometry, material recipes, transport, reflections and lighting requests. It prepares GPU requests without executing them and cannot measure rendered throughput. Table 3 partitions one future's monotonic wall lifetime into disjoint intervals. Values include scheduling/waiting within a stage, not CPU core time. Relative shares divide summed stage durations by summed preparation lifetimes; percentiles use nearest rank. Concurrent branch times and different rooms must not be summed into this partition.

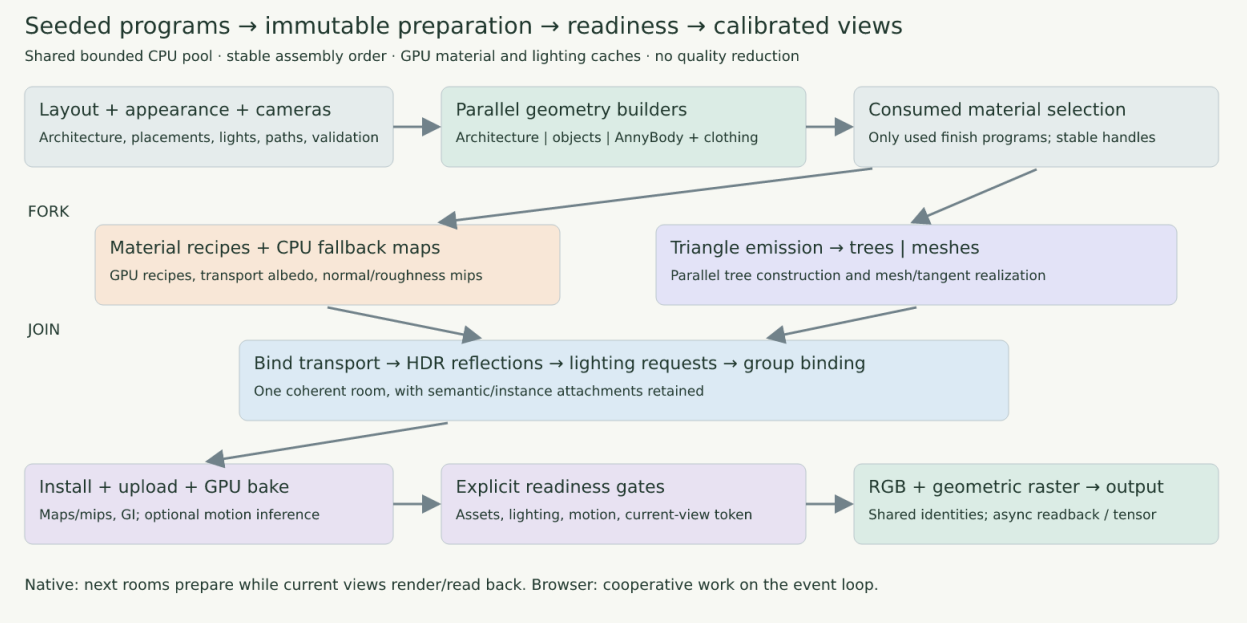


Figure 2: Production dependencies. The fork/join overlaps CPU material work with transport and mesh realization; GPU maps and lighting finish behind explicit readiness gates. Optional human inference is a separate gate. Consecutive-room lookahead spans these stages; the layout does not imply serial execution of whole rooms.

4.2 Completed capture and bounded concurrency

Each native capture run regenerates rooms and delivers three 512×512 views at one timestep in RGB, depth, position, normal, semantic and co-visibility modes. Full native shadows, SSAO, GPU materials and 256-ray/probe GI remain enabled. The default three-room lookahead is used. The streaming denominator includes validation, diagnostic JSONL writes and asset/sample destruction: preparation can advance during these operations. No image compression, dataset encoding or Burn tensor handoff is measured. Capture latency covers request to complete sample; it differs from streaming wall cost. GPU timestamp instrumentation is disabled in throughput runs.

Capture run 1 excludes 4 warm-up rooms; setup took 2.14 s and the first room took 9.97 s. The measured stream delivered 2.02 views/s; capture spans occupied 99.58% of stream wall time. These spans overlap future-room work. Main-world installation records deferred ECS commands and asset insertion, not their later execution or upload.

Preparation phase	Mean (ms)	Median (ms)	p95 (ms)	Share (%)
Layout and appearance	2.02	1.11	4.95	0.15
Camera paths and overlap	28.57	20.61	76.01	2.07
Layout validation	4.95	3.52	12.86	0.36
Motion planning	0.00	0.00	0.00	0.00
Geometry assemblies	70.19	48.34	194.05	5.09
Consumed material selection	0.11	0.10	0.21	0.01
Materials / transport / meshes	1013.12	1052.87	1474.68	73.49
Transport material binding	4.99	4.09	9.90	0.36
HDR reflection environment	211.05	172.62	461.08	15.31
Lighting bake preparation	40.13	39.60	75.99	2.91
Serial mesh tail (Wasm)	0.00	0.00	0.00	0.00
Render group binding	0.18	0.17	0.37	0.01
Scheduling and bookkeeping	3.26	2.52	8.13	0.24
Total	1378.58	1492.43	2052.31	100.00

Table 3: CPU preparation with production GPU material recipes, three square-aspect cameras and native Auto quality. The material/transport/mesh row counts the union once. Asset installation, actual GPU bakes, rendering and readback are downstream.

Workload	Rooms	Wall rooms/s	Median (ms)	p95 (ms)
CPU preparation, 1 in flight	32	0.723	1492.4	2052.3
CPU preparation, 4 in flight	32	0.783	5018.4	9684.8
Completed capture, run 1	32	0.672	1061.7	4073.5
Completed capture, run 2	32	0.669	1066.7	4177.3

Table 4: Different denominators: CPU-only preparation emits immutable asset plans; completed capture emits all requested planes. Repeated capture runs expose shared-machine variability. Neither rate includes downstream training.

Capture run 2 excludes 4 warm-up rooms; setup took 1.66 s and the first room took 9.01 s. The measured stream delivered 2.01 views/s; capture spans occupied 99.59% of stream wall time. These spans overlap future-room work. Main-world installation records deferred ECS commands and asset insertion, not their later execution or upload.

4.3 Hot spots and next optimization targets

The largest measured preparation windows are Materials / transport / meshes, HDR reflection environment and Geometry assemblies (Figure 3). They are observations of this workload, not universal cost ratios. Within the parallel asset window, mean material preparation is 993.5 ms, geometry transport construction is 790.9 ms, and mesh/tangent realization is 707.9 ms. Only ordered triangle emission precedes meshes; tree construction and mesh/tangent realization then overlap each other and material work. Geometry builder service times are separately retained by architecture, object and person categories; their sums are not a CPU-utilization measurement.

Transport construction records 44.3 ms mean triangle emission, 297.2 ms probe-tree construction and 449.2 ms shadow-tree construction. These disjoint transport stages overlap other branches and are not added to the wall partition. Cached bounds and centroids preserve median splits, equal-distance ties and the original triangle order; temporary compact records are discarded after construction.

Priority candidates are reducing triangle-staging and BVH-construction memory traffic, improving scheduling of the existing parallel transport subtrees, and GPU radiance tracing for reflection environments. Cube faces for all room probes already share immutable light inputs and execute as a bounded batch; convolution runs on the GPU, while CPU ray traversal remains a separate cost. Camera visibility searches need independent rejection/attempt profiling before changing their budget: reducing the search budget can bias the camera distribution. Geometry assembly is separately measured, so finer-grained person/architecture scheduling should be pursued only when it is on the critical path. Any move of map or reflection work to the GPU must retain deterministic inputs, normal/roughness mip correctness, annotation alignment and measured completed capture throughput while training competes for the device. Pool expansion can increase contention; more workers are not automatically faster. Broad caching keyed only by material category would reduce procedural diversity and is not an acceptable optimization.

Supported native headless captures pin current scene, time, cameras and exposure before rendering. Current-pose cluster depth is primed before Bevy assigns lights. Copies are admitted together only after the actual render, current assets/pipelines and all camera attachments qualify; unsupported temporal histories retain conservative settling.

Render-preparation, GI/material readiness, settling and readback event-loop buckets are retained in the receipt. A bucket names the blocker observed before an update; it is not an exclusive CPU or GPU execution timer. Asynchronous GPU diagnostics can arrive late and contain nested intervals, so they are not summed into per-room GPU time. A GPU timeline profiler is still needed to attribute submission gaps, uploads and individual kernels. This study does not measure isolated GPU utilization, device-to-device tensor throughput, motion inference or browser latency. Wasm compilation checks preserve the path; browser timing requires a separate hardware-WebGPU run.

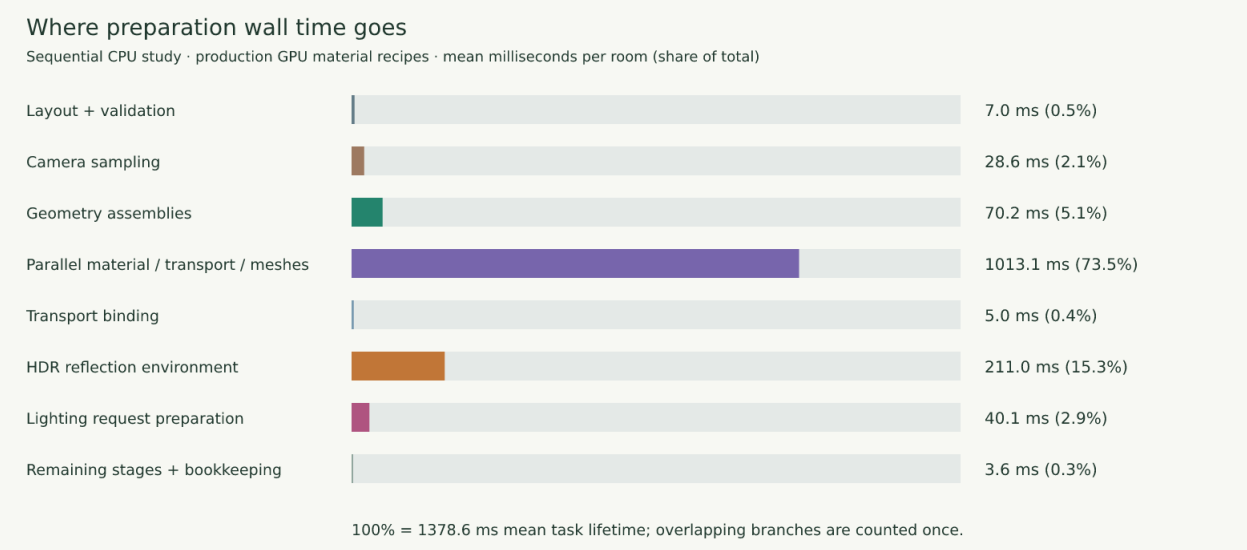


Figure 3: Mean per-room CPU preparation wall time and its relative partition for the sequential CPU study. Colors separate the major phases; the parallel window includes scheduling and all concurrent branches, exactly once. This is not a utilization chart.

Native archive consumers can optionally admit complete physical-room transactions through cumulative capture credits. Admission waits run outside the capture mutex and capture watchdog; dispatched CPU work and the bounded archive writer can finish while new idle GPU work is withheld. A permit covers every requested camera, timestep and modality. Cancellation drains admitted exports, and each requested room receives one terminal publication or failure receipt. Preparation wall partitions, overlapping branch service durations, prefetch counters, capture time, export time and demand-wait time are recorded separately. These mechanisms do not establish a concurrent-training speedup by themselves; training-update throughput and input starvation require qualification with the actual consumer.

5 Architectural evaluation

5.1 Protocol and architectural audit

This cohort was captured with bevy_zeroverse 0.44.2, generator 51, on 512 consecutive seeds (0–511), with 4 cameras per room, furnishing density 0.65 and static human density 0.25. Every audited room passed the recorded layout, support and camera-clearance checks. The first 32 rooms were rendered without appearance filtering at 640×400, with 2 trajectory endpoints: 256 camera/time identities in six aligned modes. The audit contains 512 unique structural signatures after centimetre quantization; this bounded diagnostic does not estimate the total procedural space or learning utility.

Built feature	Rooms	Fraction
Arched portals	54 / 512	10.5%
Chamfered corners	178 / 512	34.8%
Exterior cut-in	403 / 512	78.7%
Interior pillars	295 / 512	57.6%
Mezzanine	22 / 512	4.3%
Oblique walls	497 / 512	97.1%
Raised floor	76 / 512	14.8%
Sloping ceiling	354 / 512	69.1%
Sunken floor	61 / 512	11.9%

Table 5: Architectural features in the complete consecutive population. Features co-occur; fractions do not sum to 100%.

5.2 Geometry, calibration and displayed captures

Footprint areas span 34.49–413.71 m^2, with roof pitch 0.00–11.32°. Figure 4 and Figure 5 pair native RGB with the actual serialized envelope plans and sections. Camera arrows indicate direction, rather than frusta. Solid furnishings are footprint proxies; the sections show roof and floor levels and omit furnishings, pillars and partitions. Structural feature examples are selected independently of rendered appearance; all rooms remain available in the contact sheet and explorer. Metric constraints are not building-code certification.

The page presents all 32 rooms, all four cameras and both endpoints in one explorer, with 1536 channel images. RGB, depth, normals, semantics, positions, co-visibility, camera calibration and plans share the same room/time identity. RGB preserves renderer tone mapping and sRGB encoding without per-image exposure correction or cropping. All previews use lossless WebP. Depth uses Turbo (warm near, cool far) with a fixed logarithmic 0–15 m display scale, colorized from the 8-bit axial-depth/15 m preview; values beyond 15 m are clamped and background is black; normals store $(n+1)/2$, and positions are affine-normalized by the annotation AABB and clamped. Semantic colors are preserved exactly. These previews are not float32 training labels. This gallery uses the surface glass policy: geometric annotations treat panes as first surfaces. The optional through-glass policy traces straight geometry behind transmitting panes while frames and opaque frit remain occluders. Neither policy describes the reflected or refracted optical path in RGB.

The native capture protocol uses Auto quality, shadows, SSAO and approximate diffuse GI with 1024 primary rays per probe. Renderer identity and capabilities are recorded in every calibration download. All captured views passed the recorded alignment checks: maximum per-view depth/position p99 error 2.861 μm, reprojection p99 0.00050 pixels and normal-length error 2.384e-7. Every view contains at least 8 semantic classes. 19/19 rooms with humans in the camera's primary functional zone show person pixels; 6/25 rooms with humans anywhere in the main envelope never show person pixels. These are class-level observations, rather than instance-level visibility guarantees.

Scene construction through capture/export takes 2.42–6.74 seconds per room (median 4.22), excluding application startup. This is a bounded capture run, not a sustained-throughput, unlimited-process memory-stability or photographic-realism qualification.

5.3 Co-visibility in every captured room

Production co-visibility accompanies every RGB and geometry view in this cohort. Each valid source pixel stores an exact uint16 membership mask of other capture cameras that see its first surface, with the source bit excluded. One deterministic additive legend is shared across all views. The unified explorer supports all memberships and individual-peer overlays; the selected peer's own tile stays RGB.

Across every room and endpoint, 81.9% of valid source-pixel observations are visible to at least one peer. This pixel-weighted statistic pools 65536000 valid source-pixel observations; it does not count unique 3D points. Exact downloads include 16-bit grayscale membership PNGs, separate 8-bit 0/1 validity PNGs, calibrated transforms and the ordered camera legend. Zero membership with valid=1 is an unshared surface; zero membership with valid=0 is background. Co-visibility and other annotations were captured together, rather than joined from separate runs.

5.4 Matched camera and appearance qualification

A separate prespecified factor sweep covers 2 geometry families and 14 settings (120 RGB images). Most variants use two cameras; the connected-graph case uses four. Every variant captures two timestamps. Room geometry hashes are invariant across each family's camera, material, detail, illumination and exposure interventions. People are disabled in this sweep. The receipt binds the actual native executable, source identity, target, feature flags and locked package versions. All cases are retained, including failures; no model inference filters pairs. The project page provides the complete matched annotation gallery and machine-readable receipt.

Captures export a full row-major pixel-space K, image dimensions, half-pixel centers, a versioned pinhole lens contract with optional seedable optical-center displacement, normalized playback progress, eased trajectory progress and optional explicitly assigned seconds. Unassigned physical time remains unknown. Short handheld paths sample local metric translations and yaw/pitch/roll increments independently of a moving look-target, retaining swept collision checks. Requested high/low/zero proxy-overlap strata are selected before placement retries. Their all-ray proxy estimates are stored separately from rendered directed overlap over valid source pixels. A proxy-negative pair can retain a small rendered overlap. The same off-center projection drives RGB, geometry, visibility, flow, frustum culling and exported calibration. Lens distortion and rolling shutter remain gated research extensions. An optional stable connected-overlap graph permits peer chains instead of requiring every view to share camera zero. Measured overlap on the exact primary region is exported separately from whole-frame overlap, so distant exterior context cannot inflate the primary-room diagnostic.

Table 6 reports directed rendered pair means, primary-room graph component counts at a 10% bidirectional overlap threshold, pixel-weighted median triangulation angles, and the maximum per-view 99th-percentile self-reprojection error. Angle statistics omit pairs with no shared primary geometry and retain their null values in the receipt. There are 14 exact duplicate RGB images: the baseline/overlap interventions deliberately reuse the reference camera. These counts measure exact repetition in this matched study, not perceptual redundancy across a training population. Sensor white balance, blur, noise and optional JPEG are separate seeded RGB-only export transforms with individual operator regressions; they are not applied in this render sweep. This bounded check does not establish photographic realism, a calibrated sensor model, a fitted training mixture or downstream transfer.

5.5 Additional rendered populations

An independent current-source qualification covers 512 distinct rooms and 3072 camera/time views. All requested captures and aligned annotations are checked before the receipt is emitted. The table keeps each sampling population separate; the selected strata are not an IID estimate of ordinary generation. The interactive gallery above remains its own four-camera protocol. The compact downloadable receipt retains protocols, selected seeds, checks and all local input hashes; it does not bundle all training images.

5.6 Population distributions and publication contract

Room and object counts retain zero-instance rooms. Camera and photometric histograms state their own sample denominators. Placement heatmaps use 33 path samples per camera and normalize each panel independently. Feature shortcuts do not change population denominators. A seed's bit width and continuous parameter ranges do not establish

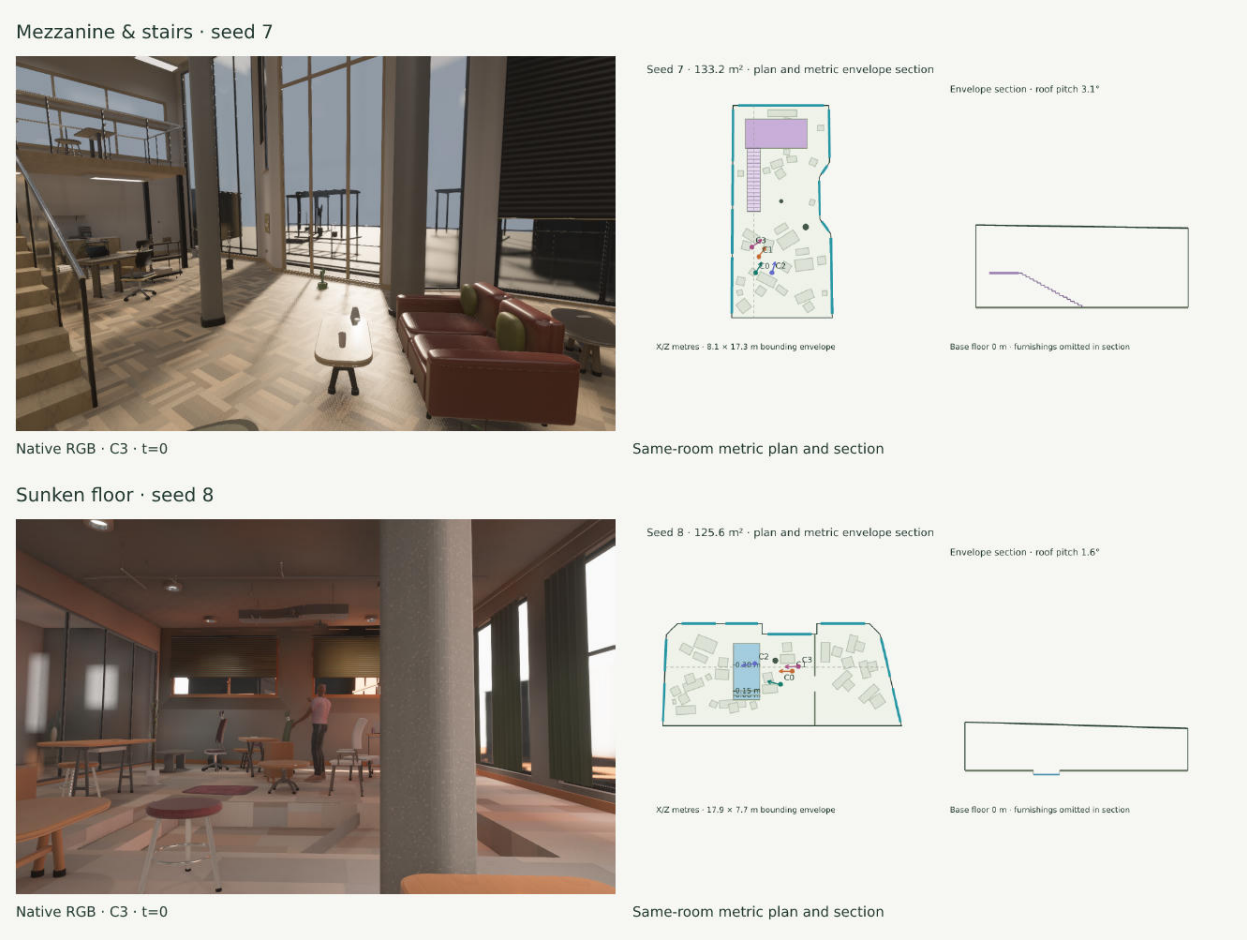


Figure 4: Native RGB and metric plans of structurally selected rooms. Raised and depressed levels, decks and stairs are generated geometry. Cyan marks glazing; orange/blue denote raised/sunken floors; violet denotes mezzanines and stairs.

Factor	Shared (%)	Base (m)	Components	θ_{50} (°)	Reproj. (px)
reference	65.13	3.082	1.00	35.1	0.00009
portrait	55.90	2.139	1.00	19.0	0.00008
principal point	63.49	2.991	1.00	27.5	0.00009
portrait principal point	59.24	2.139	1.00	19.6	0.00010
connected graph	51.12	2.187	1.00	26.4	0.00012
wide baseline	47.75	4.548	1.00	45.8	0.00007
handheld	51.59	3.581	1.00	37.2	0.00006
low overlap	14.27	2.117	1.50	8.0	0.00020
no proxy overlap	0.00	2.827	2.00	–	0.00033
material seed	65.13	3.082	1.00	35.1	0.00009
low detail	65.13	3.082	1.00	35.1	0.00009
lighting seed	65.13	3.082	1.00	35.1	0.00009
dim illumination	65.13	3.082	1.00	35.1	0.00009
exposure	65.13	3.082	1.00	35.1	0.00009

Table 6: Prespecified factor checks on matched room geometry. Directed overlap uses valid source pixels; the zero target refers to placement proxies.



Figure 5: Native RGB paired with the corresponding envelope programs. Floor levels, portals, polygonal outlines and apertures vary; features can be outside a particular camera view.

Cohort start	Rooms	Views	Shared (%)	Min. classes
0	256	1536	74.49	5
20000	128	768	73.30	5
40000	128	768	74.59	7

Table 7: Rendered populations. Shared means valid source pixels visible in any peer, pooled within each cohort; it is not unique surface area. Seeds and selection policies are retained in the receipt. No downstream model inference filters the samples.

Cohort start	Connected / groups	Median baseline (m)	Median angle (°)
0	512 / 512	2.16	19.75
20000	256 / 256	2.24	22.75
40000	256 / 256	2.10	20.25

Table 8: Measured primary-room reconstruction geometry. An edge requires at least 10% shared first-surface pixels in both directions. Groups count camera sets at each time, not independent rooms. Baselines weight each pair equally; angle values are medians of pair medians from 0.5-degree pixel-weighted source-view histograms. Empty pairs remain in the receipt and have no angle. The primary region excludes neighboring rooms and exterior context.

ten-million-sample diversity; correlations, placement constraints and rendering affect distinguishable outcomes. Frozen SigLIP2 embeddings support separate redundancy audits, but no embedding-rank or downstream-learning result is claimed here.

The project page, paper, figures and calibration archives are built by a single Rust publication pipeline from completed capture manifests. All displayed cohorts must match the current compiled generator identity, which binds crate version, geometry grammar and source/dependency hashes. The publisher validates scene/camera/time identities, every requested channel, exact membership masks, annotation checks and population denominators before staging artifacts. The release/deployment gate checks input and output hashes and rejects historical cohorts. Author-written method text and measurements have distinct roles; publication attestation does not imply photographic or downstream validation.

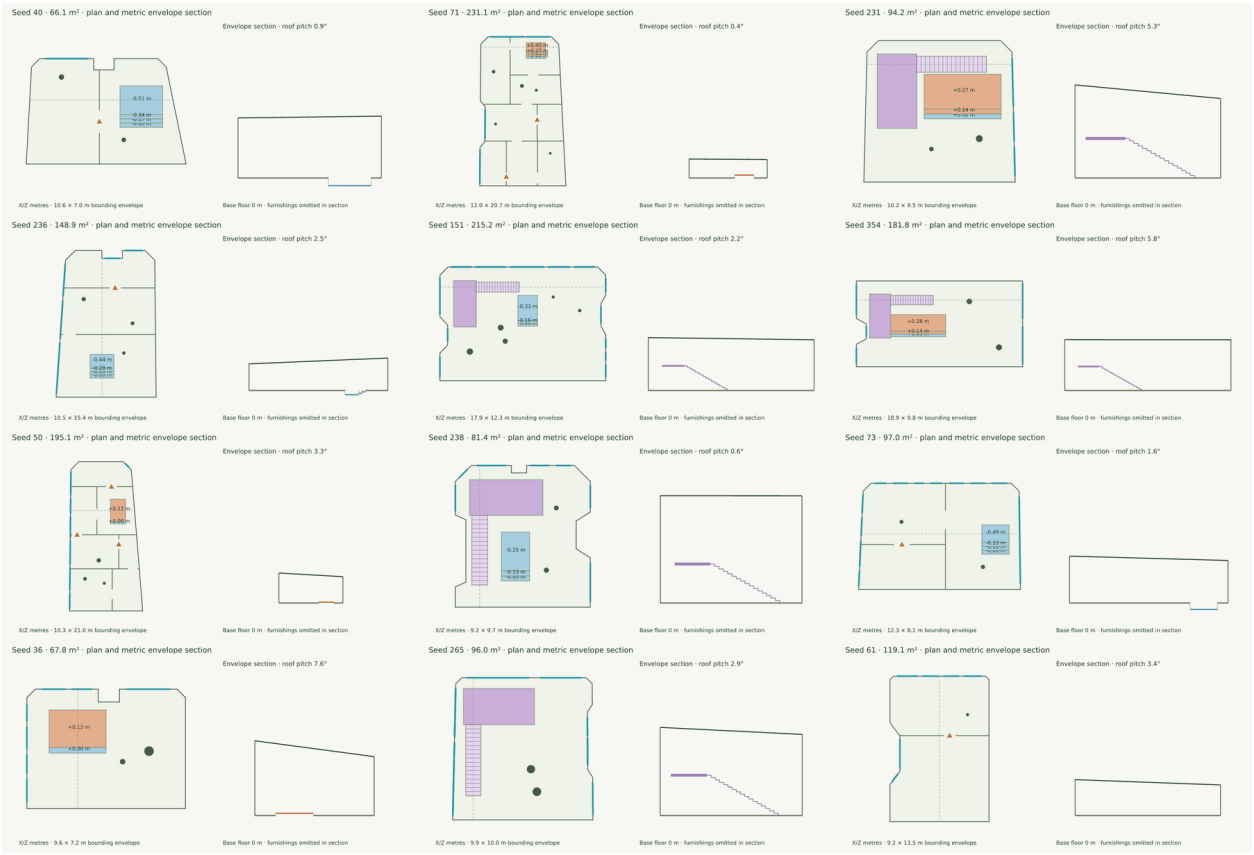


Figure 6: Twelve metric envelope programs selected by architectural feature coverage from the full consecutive population. Plans are diagrams, not twelve additional RGB renders.

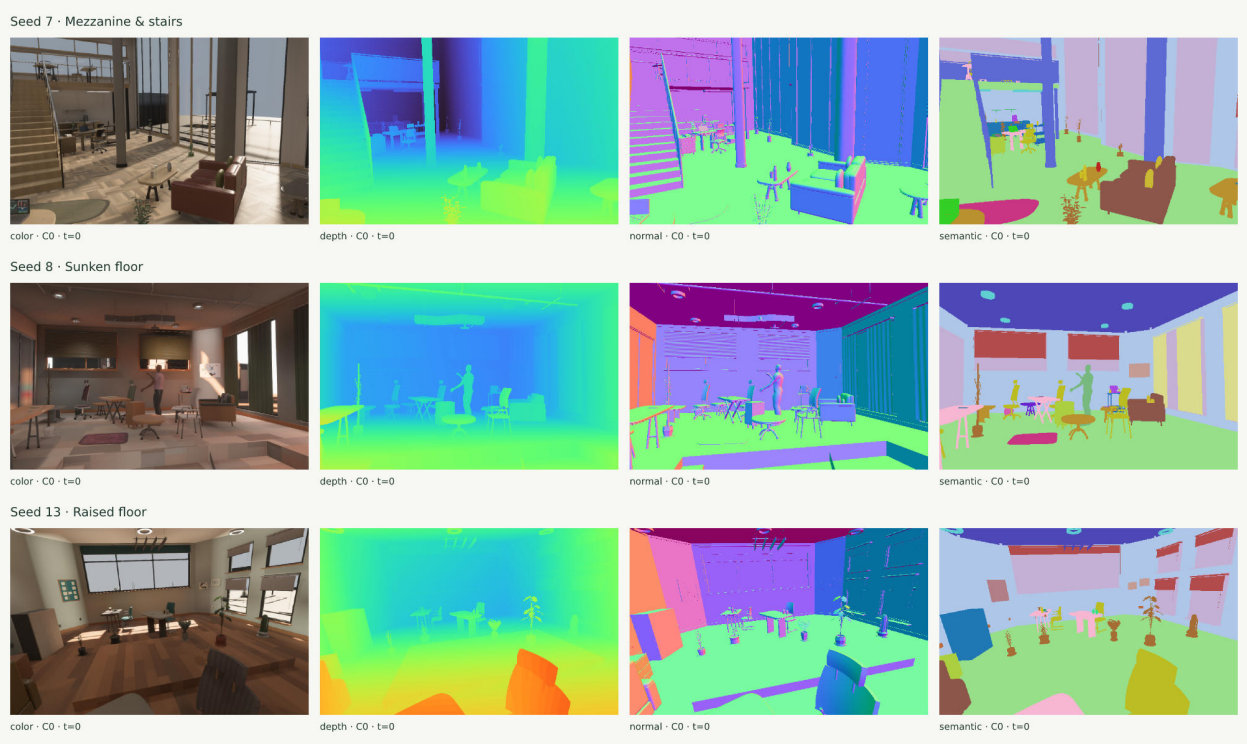


Figure 7: Matched RGB, Turbo depth, view-space normals and semantic display previews, camera 0 at t=0. All six annotations, camera identities and endpoint times remain accessible in the explorer.

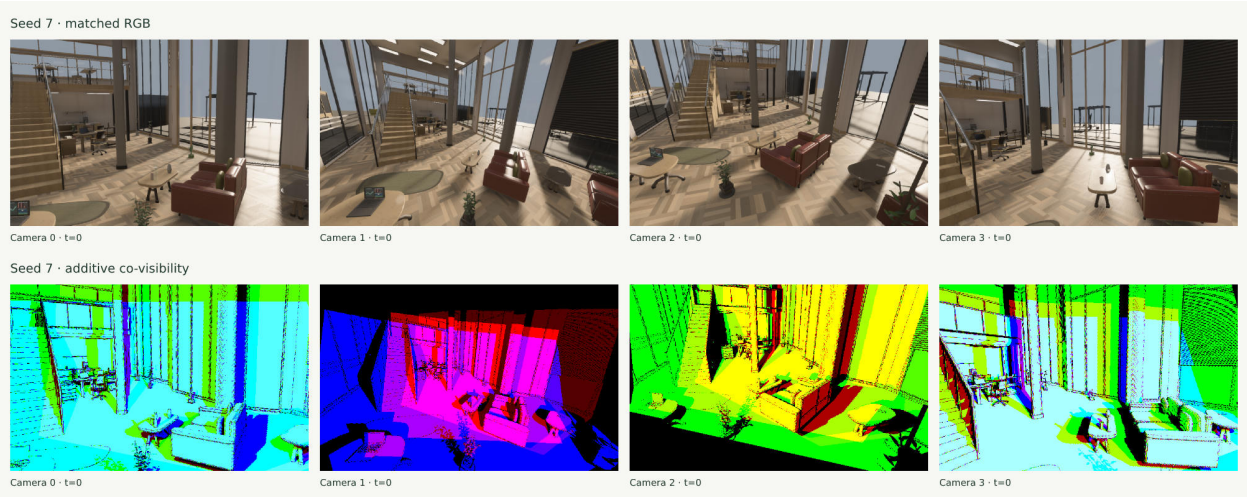


Figure 8: Seed 7 at t=0: four native RGB views above their production co-visibility annotations. Columns retain the same camera/time identity. All captured rooms expose this annotation in the same explorer.

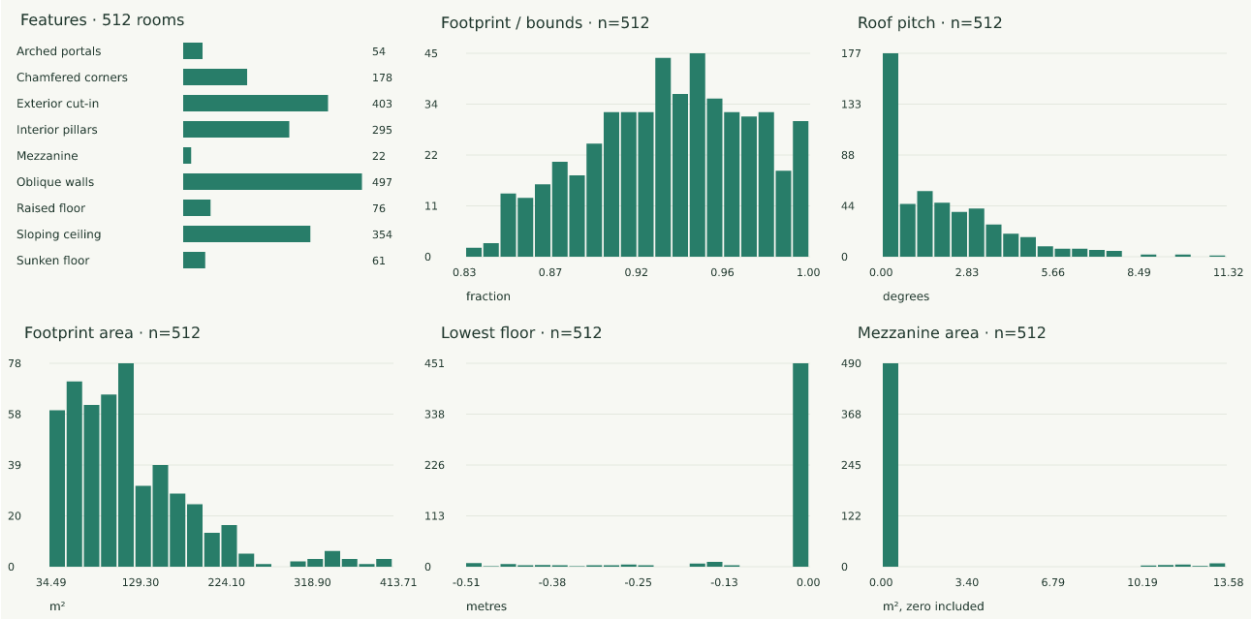


Figure 9: Architectural feature counts and continuous parameter distributions from the complete population; absent features are retained.

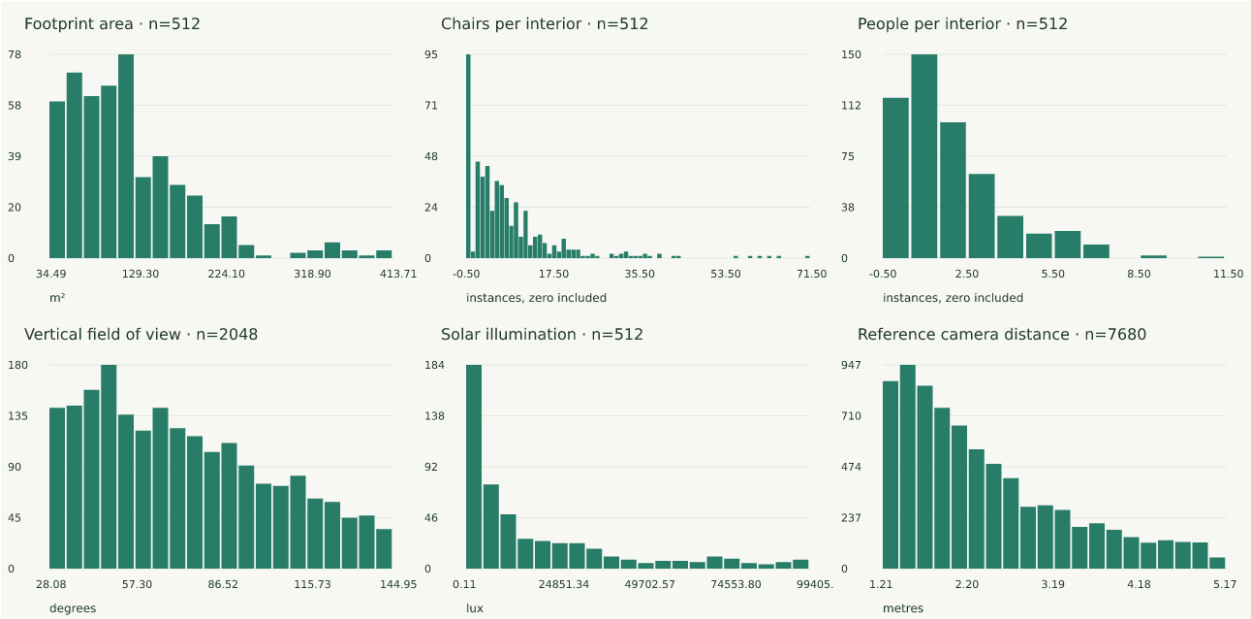


Figure 10: Furnishing, lighting and camera variation in the same population. Every panel states its denominator.

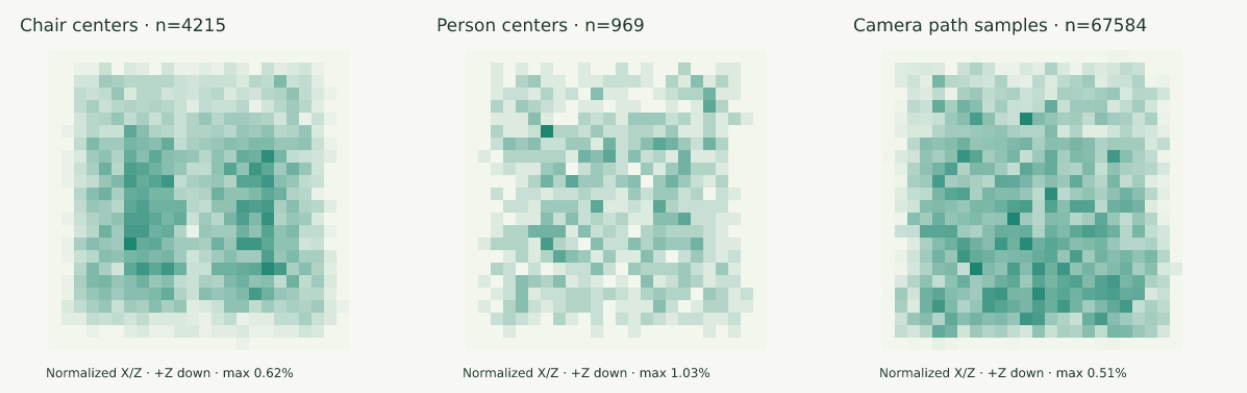


Figure 11: Chair centers, human centers and camera path samples in normalized room X/Z coordinates, with +Z down. Each panel reports its own sample count and color scale.

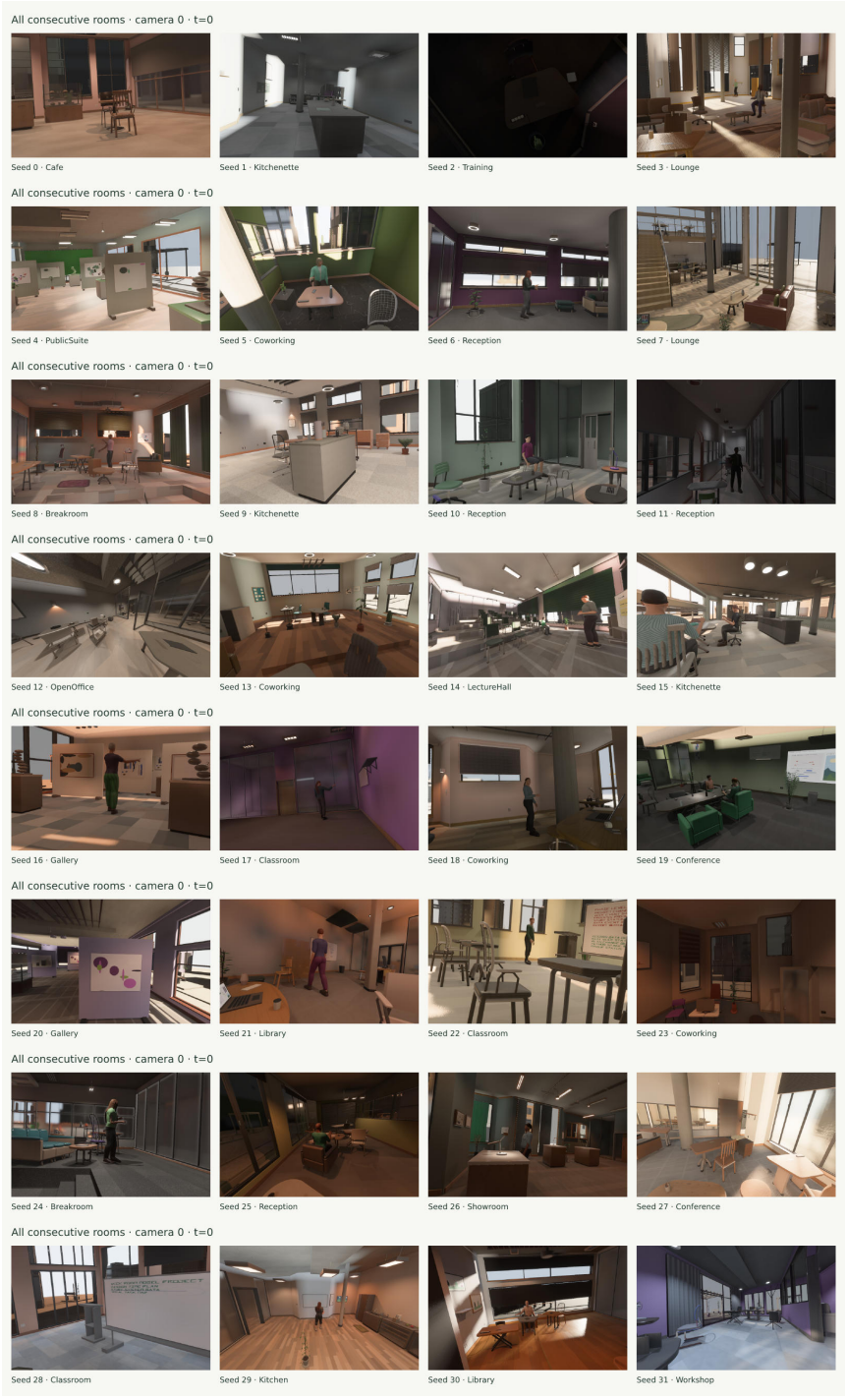


Figure 12: Every consecutively rendered room, camera 0 at t=0, including dark and sparsely furnished rooms. All four cameras, endpoints and six modes remain in the project-page explorer.

6 Camera-count and baseline audit

The current sampler was evaluated on 64 consecutive matched rooms (seeds 0–63), 1–12 cameras and baseline values 0, 0.5 and 1 at aspect ratio 1.60. Geometry, people and appearance were unchanged between requests; complete camera sets were resampled, so different counts are not necessarily nested prefixes. 0 of 2304 requested sets were rejected by the bounded sampler; all failures remain in the downloadable report. Camera-center and optical-axis measurements use 33 synchronized path samples. Reference-edge overlap uses the production 13×9 first-surface proxy at five times, rather than rendered pixels.

At one camera the baseline imposes no pair constraint. Increasing camera count does not increase the minimum nonreference pair separation: the same $0.04 + 0.46b$ metre threshold applies. Group covariance spread constrains overall anisotropy, not every nearest neighbor. Visible-region target variation samples reference-visible surface patches and ranks feasible candidates for count-aware nearest-peer spacing, optical-axis separation and additional primary-room proxy surface cells. The soft distance scale is room diagonal divided by the square root of camera count. Temporal paths can remain related deformations of an existing route; varied gaze is not independent route topology. There is no explicit 90-degree optical-angle cap.

b	Cameras	Accepted	NN (m)	Max angle (°)	Targets (m)
0.0	2	64/64	0.21	1.9	0.03
0.0	4	64/64	0.14	4.0	0.04
0.0	8	64/64	0.11	5.5	0.06
0.0	12	64/64	0.09	6.5	0.06
0.5	2	64/64	2.01	28.1	0.21
0.5	4	64/64	1.35	50.4	1.25
0.5	8	64/64	0.99	75.0	2.43
0.5	12	64/64	0.83	85.4	2.92
1.0	2	64/64	4.05	73.0	1.64
1.0	4	64/64	2.60	121.9	4.47
1.0	8	64/64	1.42	160.3	6.05
1.0	12	64/64	1.13	167.1	6.60

Table 9: Current matched placement audit. NN is the pooled camera-level initial nearest-neighbor median; angle and target diameter are room-level medians. Failed sets have no placement measurements. These denominators differ from rendered pixel overlap.

Similar absolute trajectories have RMS center separation below 0.15 m and RMS optical-axis difference below 5°. Shape similarity instead removes each start position and normalizes the RMS displacement residual by the larger RMS travel, with a diagnostic threshold of 0.25; static pairs are excluded. Close stereo views can intentionally satisfy the pose threshold. A common-axis least-squares residual measures convergence of initial optical-axis lines; parallel or ill-conditioned lines have no reported focus. Neither statistic proves perceptual repetition or reconstruction utility.



Figure 13: Camera-count distributions at three baseline settings. Room-level viewing-angle/target statistics and camera-level spacing use their own denominators. Repeated path shapes are measured separately from repeated absolute poses.



Figure 14: Baseline 1 camera-path heatmaps for 1–12 views: 33 synchronized samples per camera, primary-room bounds normalized per room and +Z down. A common probability color scale is used. These show positions, not reconstructed-surface coverage.

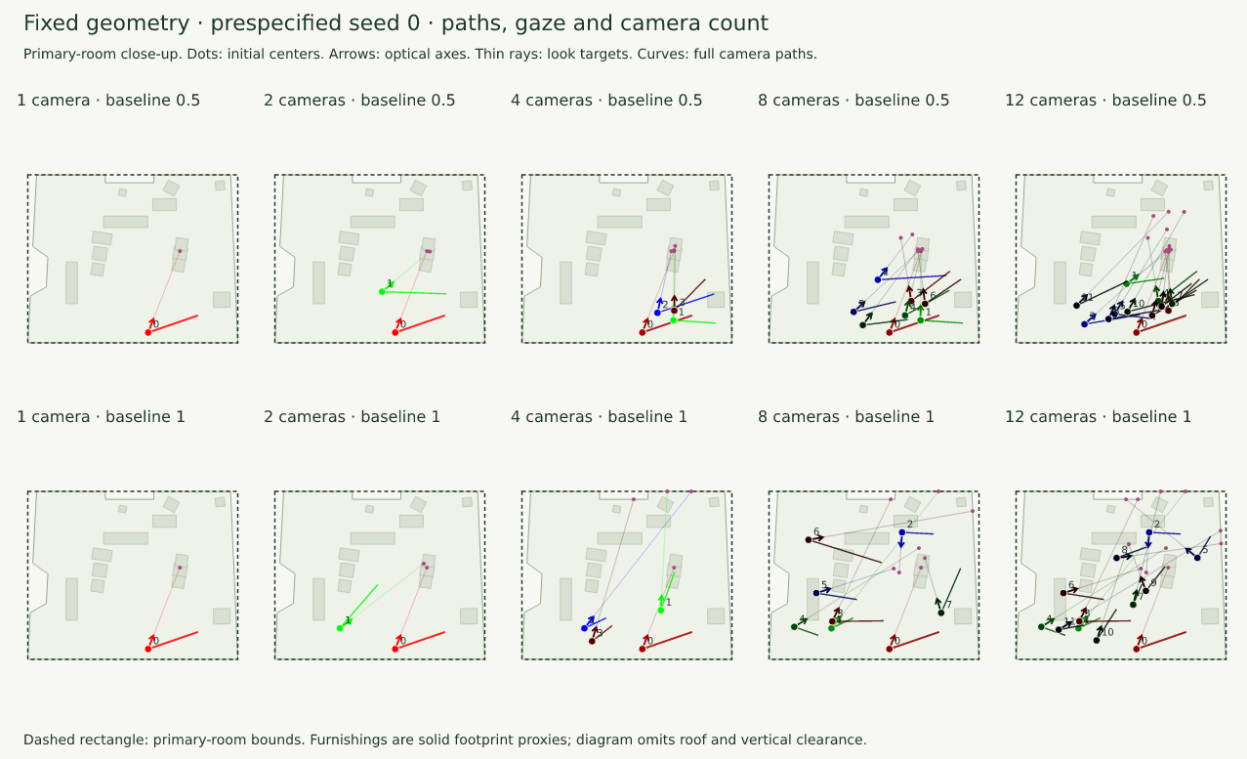


Figure 15: Prespecified first room, fixed geometry and people, default/wide baselines and increasing camera counts. Full paths, initial optical directions and look targets expose the shared-content proposal. Rejected requests are retained.

6.1 Native rendered overlap across camera counts

The rendered check uses 2 prespecified matched room families, 12 configurations and 312 camera/time views: counts 2, 4, 8 and 12 at all three baselines and both trajectory endpoints, with 320×200 images, static human density 0.25 and native shadows, material synthesis and diffuse GI retained. The executable and every channel are hash-bound. Rendered centers and optical axes match the CPU audit within 10^{-4} m and 10^{-5} per direction component. Table 10 separates mean directed pair overlap from mean source-view visibility in any peer, and reports the smallest reference-edge overlap over the sampled rooms/times. These do not measure unique scene-surface coverage.

b	Cameras	Pair (%)	Any peer (%)	Min ref. (%)	Components
0	2	87.1	87.1	75.4	1
0	4	87.0	94.2	67.7	1
0	8	85.7	98.3	67.1	1
0	12	85.6	98.8	67.1	1
0.5	2	66.7	66.7	40.2	1
0.5	4	53.3	80.1	39.1	1
0.5	8	55.6	91.8	38.6	1
0.5	12	55.9	94.6	37.2	1
1	2	41.3	41.3	28.3	1
1	4	43.8	78.8	22.8	1
1	8	40.0	88.8	18.1	1
1	12	39.2	92.2	18.1	1

Table 10: Native first-surface co-visibility; glass follows the solid annotation convention. Components is the largest number of connected components when an undirected edge requires at least 5% shared pixels in both directions. The two room families are the independent units; camera pairs and endpoints are correlated.

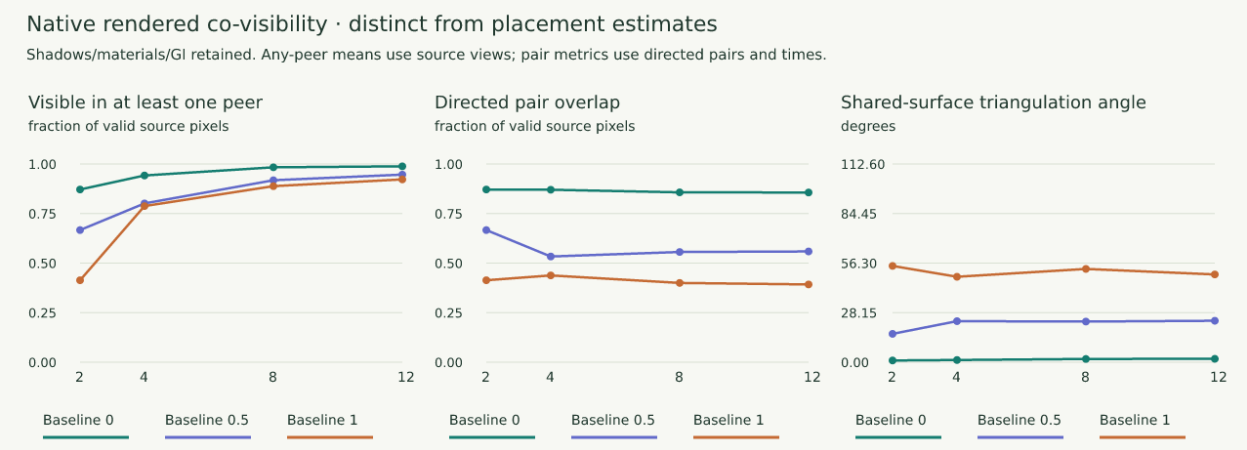


Figure 16: Measured native visibility and triangulation at increasing camera counts. Any-peer visibility is averaged over source views, other means over directed pairs. More peer views increase any-peer visibility without proving broader unique surface coverage.

6.2 Visual effect of camera baseline

Each column resamples a complete four-camera group over identical geometry, people, materials and lighting. Camera indices identify group members, rather than fixed physical viewpoints across columns. These are the first two prespecified native families, with every view retained at t=0 and no per-image exposure correction. They contribute no extra rooms to the population denominator.



Figure 17: Fixed room 0: narrow, default and wide baseline settings (0, 0.5 and 1) from left to right. Rows retain all four native RGB views at 320×200. Perspective and shared content change with the camera policy; scene content is invariant.

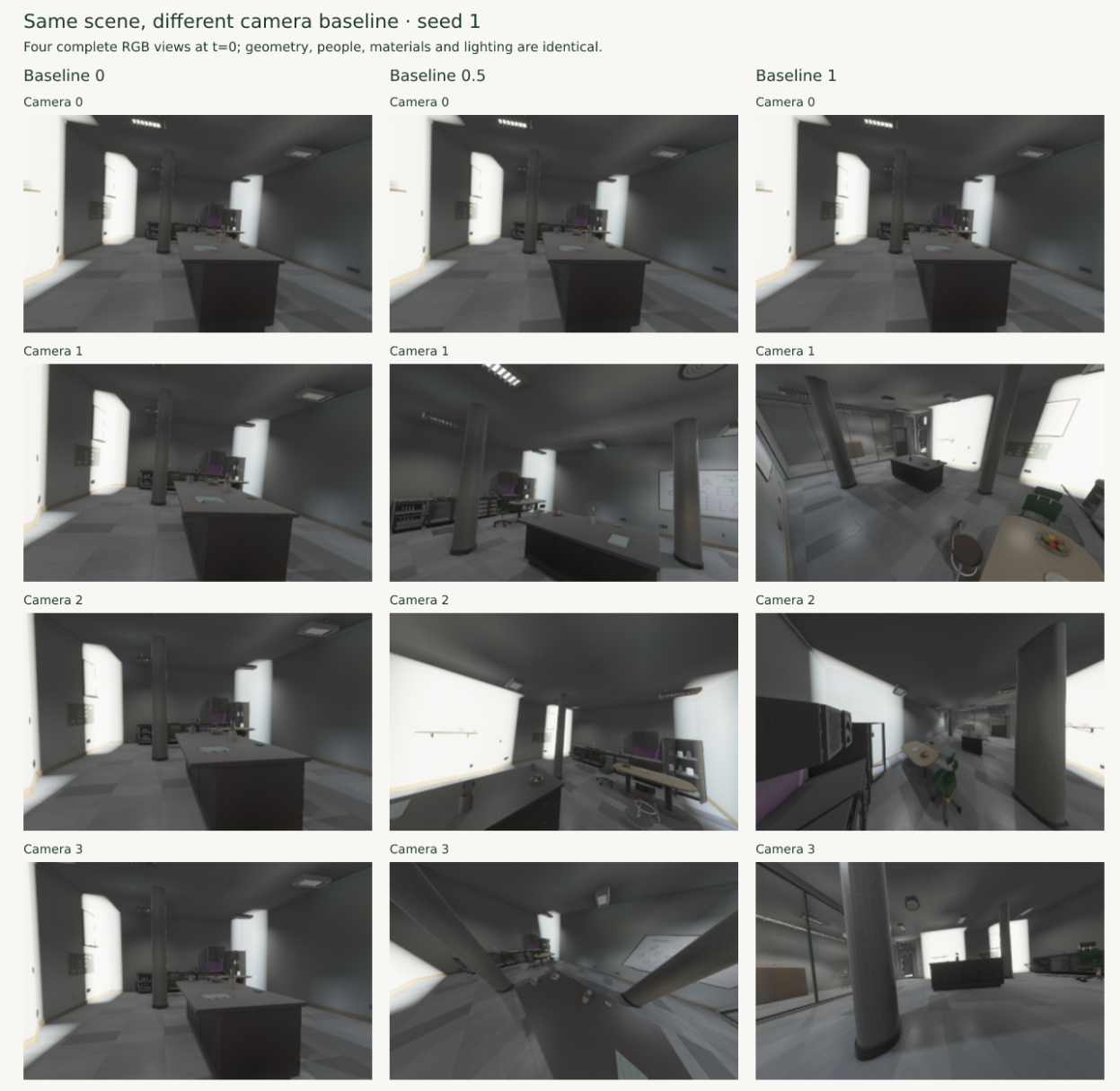


Figure 18: Fixed room 1: narrow, default and wide baseline settings (0, 0.5 and 1) from left to right. Rows retain all four native RGB views at 320×200. Perspective and shared content change with the camera policy; scene content is invariant.

7 Object geometry diversity

7.1 Fixed-envelope protocol

We evaluate all 100 furnishing/prop categories on 256 consecutive construction seeds each: 25600 triangle assemblies. Each category holds its metric placement envelope, orientation and finish seed fixed. This isolates internal construction from changes in object size, pose or color; it deliberately does not measure the full joint scene distribution. The cable fixture varies a controlled centerline; actual endpoint connectivity and collision are separate room-level checks. Humans and architecture are evaluated elsewhere.

For each mesh, six orthographic first-surface depth maps sample pixel centers at 24×24 resolution per direction. Depth is normalized by the fixed envelope extent. Pair distance is mean absolute depth difference over the union of hit pixels; missing-versus-present surface contributes one. Greedy coverage retains a sample only when its distance to every earlier representative exceeds the threshold. We report thresholds 0.005, 0.015 and 0.030, and retain seed order. These are finite-resolution, order-dependent coverage counts, not numbers of independent geometries, perceptual embedding volume, or a proof of a multiplicative increase in dataset utility. Thin features and content on nearly planar surfaces can be under-resolved; saturated categories need more samples or a finer descriptor.

All audited assemblies passed finite position/normal/UV, unit-normal, winding, nondegenerate-triangle, index, semantic-label and reserved-envelope checks (maximum allowed overrun 3 mm). This does not prove watertightness, lack of every self-intersection, or physical plausibility. Separate regression tests cover support, cable attachment, overlapping planar faces and deterministic replay. Mesh build time excludes placement, materials, uploads and rendering and is not completed-room throughput.

Six examples per category are selected by deterministic farthest-spread sampling from the same depth descriptors, beginning with seed zero. The native renderer reuses one device, with fixed cameras, exposure, lights and finish seed within each category. The project-page atlas provides RGB and normal views, every example's program inputs, all sampled parameter marginals and the complete machine-readable measurements. These selected examples are correlated draws, not one-parameter interventions; inactive conditional controls are not evidence of realized shape diversity.

7.2 Category measurements

Table 11: Current construction coverage at three distances; triangles and CPU build milliseconds are per-object medians. All categories use 256 seeds.

Category	$C_{.005}$	$C_{.015}$	$C_{.030}$	Triangles	ms
Table	255	246	225	2272	0.095
Desk	255	246	219	2184	0.079
Chair	255	252	241	4444	0.118
Sofa	256	252	176	11552	0.400
CoffeeTable	256	255	236	2200	0.062
Cabinet	135	74	32	1888	0.068
Bookcase	256	251	163	9576	0.276
Plant	256	256	256	5240	0.213
TrashCan	219	138	54	1452	0.031
Whiteboard	51	26	9	646	0.028
Display	91	47	34	132	0.006
Laptop	252	200	122	8012	0.255
Monitor	256	255	250	668	0.030
Notebook	124	76	32	626	0.023
Mug	212	97	36	1284	0.032
Books	256	232	137	2160	0.045
Rug	167	66	30	768	0.028
WallArt	15	3	2	542	0.013
Clock	159	58	9	5560	0.232
FloorLamp	256	256	256	3232	0.083
Keyboard	252	235	150	7020	0.217
Mouse	251	210	115	1040	0.060
WaterBottle	160	67	22	3104	0.065
PenHolder	256	256	256	752	0.033
Printer	171	49	15	776	0.034

Category	$C_{.005}$	$C_{.015}$	$C_{.030}$	Triangles	ms
StorageBox	188	55	18	452	0.034
CoatRack	233	205	159	3280	0.075
Bag	246	120	33	736	0.041
CoffeeCup	112	65	30	956	0.038
SodaCan	11	2	1	1752	0.045
Notepad	43	23	10	240	0.044
Pencil	59	21	9	116	0.008
Microphone	168	152	103	1076	0.023
Phone	66	20	13	200	0.008
WallOutlet	14	7	3	1252	0.030
LightSwitch	31	14	6	724	0.029
SideTable	256	250	236	2200	0.072
MobileCart	147	92	56	3348	0.092
DrawerUnit	13	3	3	1400	0.047
Bench	23	13	8	1136	0.041
Ottoman	144	73	32	1840	0.062
RoomDivider	23	17	17	540	0.029
DeskLamp	253	239	219	1514	0.032
CoffeeMaker	129	38	21	1132	0.049
Kettle	150	39	14	1628	0.034
Speaker	193	82	28	1232	0.060
Headphones	250	213	136	2624	0.065
Stapler	152	52	18	404	0.012
TapeDispenser	201	105	44	648	0.019
DocumentTray	34	23	12	2052	0.041
DeskOrganizer	8	8	8	888	0.022
TissueBox	256	201	61	1292	0.043
Vase	252	241	183	384	0.010
Bowl	218	141	68	384	0.014
DeskFan	71	35	18	4276	0.136
Projector	93	10	2	2728	0.062
ProjectionScreen	18	10	8	514	0.013
PowerStrip	4	2	1	2052	0.044
Cable	209	188	143	496	0.018
WallShelf	34	30	17	700	0.018
NoticeBoard	2	1	1	7032	0.171
WallSconce	235	174	103	344	0.041
CeilingPendant	248	222	194	1928	0.050
FloorOutlet	73	72	31	668	0.018
KitchenCounter	147	87	50	4728	0.097
KitchenSink	188	111	64	7240	0.171
KitchenCupboard	254	232	163	7352	0.220
Pantry	115	63	40	4032	0.086
Cooktop	36	14	11	1836	0.053
Range	143	81	47	6268	0.230
Oven	154	98	62	4184	0.177
Microwave	151	100	67	2400	0.079
Dishwasher	89	47	24	5952	0.124
Refrigerator	115	71	42	3904	0.085
DryingRack	83	35	13	1884	0.042
SoapDispenser	3	2	2	692	0.018
Sponge	4	1	1	216	0.007
KnifeBlock	10	4	4	1080	0.037
UtensilHolder	171	77	33	1468	0.043
PaperTowel	105	43	19	760	0.020
KitchenTowel	1	1	1	432	0.010
Dishware	137	86	51	240	0.007
CuttingBoard	1	1	1	324	0.012
FruitBowl	227	155	88	2920	0.094
Bread	1	1	1	490	0.086
Elevator	8	5	3	3076	0.084
ServiceCounter	1	1	1	972	0.024
DisplayCase	1	1	1	5250	0.253

Category	$C_{.005}$	$C_{.015}$	$C_{.030}$	Triangles	ms
TrayReturn	4	2	1	1404	0.031
DrinkDispenser	1	1	1	752	0.021
QueueBarrier	19	6	3	1280	0.047
Lectern	1	1	1	836	0.023
Locker	28	16	7	5940	0.163
TeachingBoard	1	1	1	2568	0.070
ExhibitPlinth	1	1	1	324	0.010
Sculpture	117	56	32	3148	0.068
InformationStand	1	1	1	308	0.009
ExhibitionPanel	1	1	1	1408	0.043
ExtractionHood	1	1	1	384	0.012
Banquette	2	1	1	756	0.018



Figure 19: Matched studio construction examples: Chair, Table, Sofa, Plant. All columns within each row share dimensions, viewpoint, lights and finish seed. Seed labels link to the full sampled parameter grid in the downloadable atlas. Example selection emphasizes shape spread, not frequency or image quality; the table includes every sampled seed.



Figure 20: Matched studio construction examples: Keyboard, Mouse, FloorLamp, CoatRack. All columns within each row share dimensions, viewpoint, lights and finish seed. Seed labels link to the full sampled parameter grid in the downloadable atlas. Example selection emphasizes shape spread, not frequency or image quality; the table includes every sampled seed.



Figure 21: Matched studio construction examples: StorageBox, Printer, Bag, Headphones. All columns within each row share dimensions, viewpoint, lights and finish seed. Seed labels link to the full sampled parameter grid in the downloadable atlas. Example selection emphasizes shape spread, not frequency or image quality; the table includes every sampled seed.

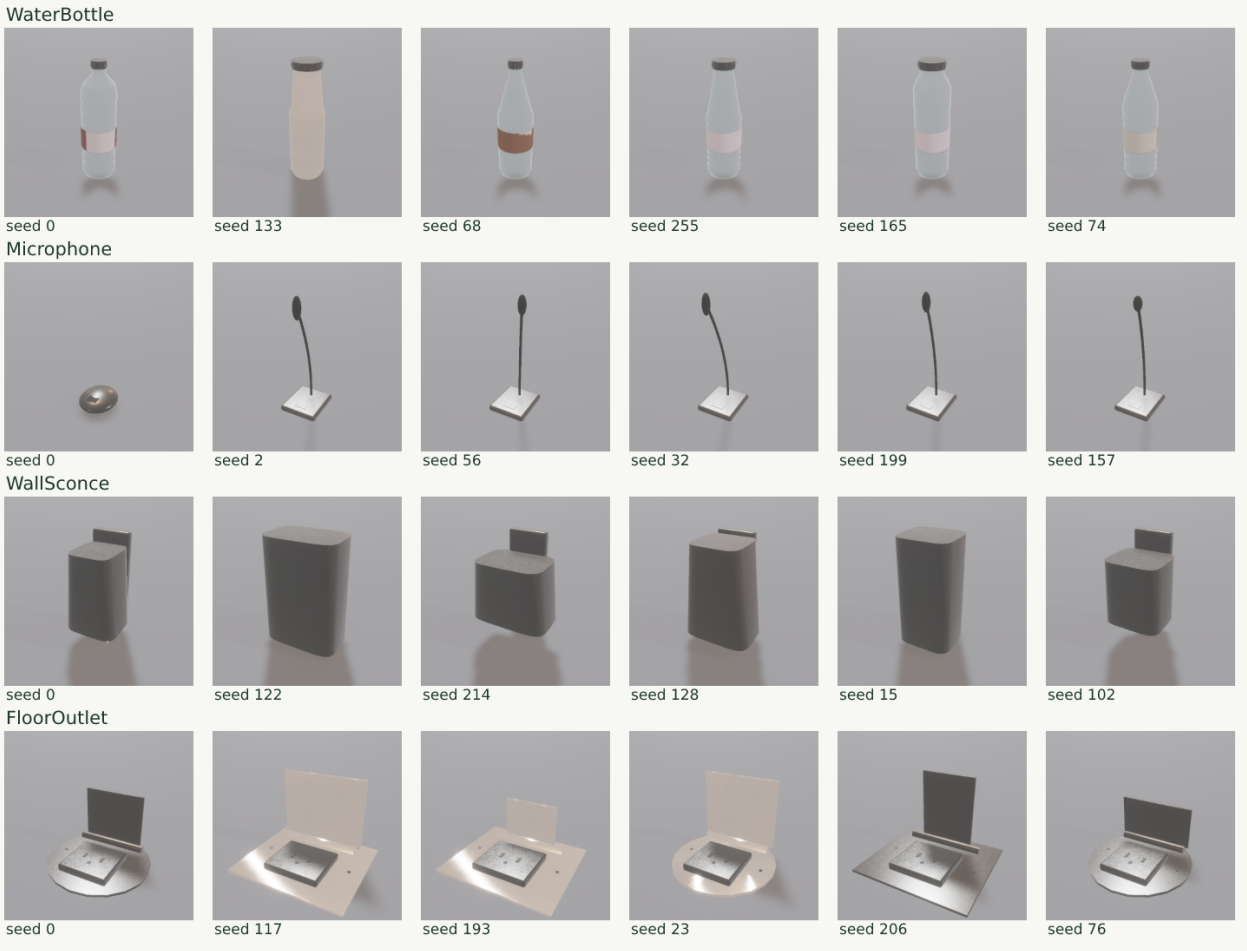


Figure 22: Matched studio construction examples: WaterBottle, Microphone, WallSconce, FloorOutlet. All columns within each row share dimensions, viewpoint, lights and finish seed. Seed labels link to the full sampled parameter grid in the downloadable atlas. Example selection emphasizes shape spread, not frequency or image quality; the table includes every sampled seed.



Construction seed	0	138	199	182	171	101
Shade height (m)	0.248	0.238	0.324	0.224	0.315	0.254
Radius / envelope width	0.402	0.434	0.372	0.313	0.429	0.401
Upper / lower radius	0.993	0.656	0.768	0.623	0.980	0.855
Stem offset / width	0.053	-0.067	0.071	0.047	-0.080	0.090
Shade sides	32	32	32	32	6	32
Support code	3	1	2	1	2	1

Figure 23: A construction parameter grid with values taken from the pictured meshes. Each lamp occupies the same $0.55 \times 1.65 \times 0.55$ m reserved envelope. Support code 0 is a disk and stem, 1 a wooden tripod, and 2 four metal legs. A 32-sided shade approximates a circular section; smaller counts produce polygonal shades. Stem offset and shade radius are jointly bounded by the reserved envelope. The bulb and light emitter use the same computed position. This grid shows correlated sampled programs, not isolated causal effects.

7.3 Sampled controls and their interpretation

The following inventory lists the sampled construction inputs recorded by each builder. Some controls are conditional on a construction choice (for example, a phone's rear lenses affect only face-down configurations); a range does not imply independent influence. Observed ranges, eight-bin numeric marginals, categorical frequencies and all seed-level values accompany the project-page grids. Conditional controls use only samples in which the control is active; their active and total denominators are exported explicitly. A small coverage count for a flat notebook, notice board or outlet is retained explicitly rather than counting different textures as different geometry.

Category	Recorded controls
Table	capsule_ blend, edge_ bevel, frame_ bend_ m, leg_ aspect, leg_ inset, leg_ radius, leg_ rake, leg_ taper, outline_ exponent, pedestal_ base_ aspect, pedestal_ base_ radius, pedestal_ radius, spider_ sweep, support, taper, top_ surface, top_ thickness, trestle_ spread
Desk	capsule_ blend, edge_ bevel, frame_ bend_ m, leg_ aspect, leg_ inset, leg_ radius, leg_ rake, leg_ taper, outline_ exponent, pedestal_ base_ aspect, pedestal_ base_ radius, pedestal_ radius, spider_ sweep, support, taper, top_ surface, top_ thickness, trestle_ spread
Chair	arm_ height, arm_ pad_ length, arm_ pad_ roundness, arm_ pad_ width_ m, arm_ support, armrests, back_ construction, back_ width_ fraction, base_ radius_ fraction, curvature, frame_ bend_ m, headrest, headrest_ curvature_ m, headrest_ height_ m, headrest_ width_ fraction, leg_ splay, lumbar, mesh_ pitch_ m, recline, seat_ crown, seat_ depth_ fraction, seat_ dish_ m, seat_ roundness, seat_ width_ fraction, shell_ thickness, shoulder_ flare, spindle_ pitch, spoke_ count, taper
Sofa	arm_ crown, arm_ rise_ m, arm_ roundness, arm_ width, back_ tilt, chaise, cushion, cushion_ crown, cushion_ dish_ m, cushion_ gap_ m, exposed_ frame, left_ return, leg_ height, module_ asymmetry, modules, pillow_ roll, pillow_ width_ m, pillows, roundness, seam_ radius_ m, seat_ height, upholstery
CoffeeTable	capsule_ blend, edge_ bevel, frame_ bend_ m, leg_ aspect, leg_ inset, leg_ radius, leg_ rake, leg_ taper, outline_ exponent, pedestal_ base_ aspect, pedestal_ base_ radius, pedestal_ radius, spider_ sweep, support, taper, top_ surface, top_ thickness, trestle_ spread
Cabinet	columns, fill, front, handle, levels, metal, open_ back, open_ fraction, opening_ radians, panel_ m
Bookcase	columns, fill, front, handle, levels, metal, open_ back, open_ fraction, opening_ radians, panel_ m
Plant	branching, crown_ skew, density, droop, leaf_ aspect, maturity, phyllotaxis, pot_ fraction, pot_ radius, pot_ taper
TrashCan	finish, lid, rim_ width, squareness, taper
Whiteboard	border_ m, marker_ count, mounts, tray, tray_ fraction, tray_ side
Display	aspect, bezel_ m, curvature_ radius_ m, lift_ fraction, panel_ m, stand, tilt_ radians
Laptop	aspect_ ratio, bezel_ m, chassis_ m, corner_ m, hinge_ radians, keyboard_ columns, keyboard_ fraction, lid_ m, port_ count, speaker_ grilles, trackpad_ fraction
Monitor	aspect, bezel_ m, curvature_ radius_ m, lift_ fraction, panel_ m, stand, tilt_ radians
Notebook	binding, cover_ fraction, opened, page_ balance, page_ crown, strap
Mug	fill_ fraction, style, taper, wall_ m
Books	binding, cover_ fraction, size_ variation, stack_ yaw, strap, volumes
Rug	clipped_ corners, rug_ exponent, rug_ taper
WallArt	border_ width, mat_ width, picture_ inset, shadow_ gap, style
Clock	bezel_ fraction, dark_ dial, digital, second_ hand, seconds, square
FloorLamp	shade, shade_ height, shade_ radius_ fraction, shade_ sides, shade_ taper, stem_ offset, support
Keyboard	deck_ fraction, key_ area_ fraction, keyboard_ columns, keyboard_ rows, keyboard_ split, row_ stagger, wrist_ rest
Mouse	button_ fraction, mouse_ crown, mouse_ roundness, mouse_ waist, trackball
WaterBottle	fill_ fraction, neck_ fraction, shoulder_ fraction, style
PenHolder	variant
Printer	construction, lid_ fraction, roundness, split
StorageBox	construction, handle_ span, lid_ fraction, roundness, taper
CoatRack	arms, branch_ height, construction, handle_ span
Bag	construction, handle_ span, lid_ fraction, roundness, taper
CoffeeCup	fill_ fraction, lid, style, taper, wall_ m
SodaCan	heel_ fraction, lid, neck_ fraction, style
Notepad	binding, leaf_ fraction
Pencil	barrel_ sides, binding, tip_ fraction
Microphone	microphone_ bend, microphone_ head, microphone_ lateral
Phone	bezel_ fraction, camera_ island, case_ fraction, lens_ count
WallOutlet	edge_ radius, finish, gangs, rocker_ angle, screws, style
LightSwitch	edge_ radius, finish, gangs, rocker_ angle, screws, style

Category	Recorded controls
SideTable	capsule_ blend, edge_ bevel, frame_ bend_ m, leg_ aspect, leg_ inset, leg_ radius, leg_ rake, leg_ taper, outline_ exponent, pedestal_ base_ aspect, pedestal_ base_ radius, pedestal_ radius, spider_ sweep, support, taper, top_ surface, top_ thickness, trestle_ spread
MobileCart	feet_ height, levels
DrawerUnit	drawer_ bias, drawers, feet_ height, handle_ fraction
Bench	backed, roundness, slats
Ottoman	feet_ height, roundness
RoomDivider	panel_ fraction, slats
DeskLamp	arm, head, shade
CoffeeMaker	belly, neck, style, taper
Kettle	belly, style, taper
Speaker	body_ taper, control_ fraction, profile_ roundness, style, support_ fraction
Headphones	control_ fraction, profile_ roundness, support_ fraction
Stapler	body_ taper, control_ fraction, support_ fraction
TapeDispenser	body_ taper, control_ fraction, support_ fraction
DocumentTray	body_ taper, sections, support_ fraction
DeskOrganizer	body_ taper, sections, support_ fraction
TissueBox	box_ height, finish, fold, lean, opening, phase
Vase	belly, flutes, neck, style, taper
Bowl	belly, flutes, style, taper
DeskFan	style, support_ fraction
Projector	corner_ m, lens_ fraction, lens_ offset, vent_ count
ProjectionScreen	screen_ drop, screen_ frame
PowerStrip	corner_ m, socket_ count
Cable	variant
WallShelf	bracket_ style, shelf_ brackets, shelf_ span
NoticeBoard	board_ divider_ fraction, board_ frame_ fraction, note_ count
WallSconce	diffuser_ fraction, sconce_ height, sconce_ taper, sconce_ width
CeilingPendant	diffuser_ fraction, shade_ fraction
FloorOutlet	lid_ height_ fraction, lid_ width_ fraction
KitchenCounter	counter_ doors, doors, drawers, front, front_ profile, handles, hardware, installation, opening_ radians, top
KitchenSink	basins, cabinet_ motion, dish_ soap, dishes, door_ style, draining_ board, faucet, faucet_ handles, front_ profile, hand_ soap, handles, opening_ radians, sponge, top
KitchenCupboard	cabinet_ motion, contents, door_ style, finish, front_ profile, handles, hardware, opening_ radians, paired, shelves, upper_ bottom_ fraction, upper_ depth_ fraction, upper_ offset_ fraction, upper_ width_ fraction
Pantry	cabinet_ motion, contents, door_ style, finish, front_ profile, handles, hardware, opening_ radians, paired, shelves, upper_ bottom_ fraction, upper_ depth_ fraction, upper_ offset_ fraction, upper_ width_ fraction
Cooktop	controls, gas
Range	controls, gas, opening_ radians, viewport
Oven	controls, installation, opening_ radians, stacked, viewport
Microwave	contents, controls, opening_ radians, viewport
Dishwasher	contents, opening_ radians, viewport
Refrigerator	contents, door_ notes, opening_ radians, paired, shelves, viewport
DryingRack	count
SoapDispenser	pump
Sponge	size_ metres
KnifeBlock	count
UtensilHolder	count
PaperTowel	horizontal
KitchenTowel	size_ metres
Dishware	stack_ count, style
CuttingBoard	size_ metres
FruitBowl	basket, fruit_ count
Bread	count
Elevator	bays, finish, opening
ServiceCounter	bays, finish
DisplayCase	bays, contents, finish, shelves
TrayReturn	contents, finish, shelves
DrinkDispenser	opening
QueueBarrier	opening

Category	Recorded controls
Lectern	finish
Locker	bays, contents, finish, opening
TeachingBoard	dimensions_ metres
ExhibitPlinth	finish
Sculpture	bays, curvature, glazed
InformationStand	dimensions_ metres
ExhibitionPanel	taper
ExtractionHood	dimensions_ metres
Banquette	bays, finish

7.4 Door and cabinetry construction

Doors form a separate architectural inventory from placeable furniture. Hinged, paired, surface-sliding, telescoping and revolving mechanisms sample continuous states, handedness, dimensions, panel construction and hardware finishes. Revolving entrances require a wider aperture and are reserved as static obstacles; the generator does not simulate automatic door operation. Arch frames have no door-leaf annotation. Cabinet fronts mix hinged, sliding and upper lift-up mechanisms; lower and upper cupboard states share a recorded construction program. Opening travel is admitted before placement. RGB, depth, normals and semantic meshes use the same geometry. Door leaves, frames and hardware form one bounding-box instance, with distinct semantic classes; cabinet glazing retains cabinet ownership and cabinet pulls have their own hardware label. The category audit checks 64 consecutive samples in each of ten categories (640 assemblies). Six illustrative samples per category are captured in aligned PBR, normal and semantic channels. They illustrate state/material coverage rather than occurrence rates. Cabinet envelopes include opening travel; consequently these are not fixed-envelope perceptual diversity estimates. Triangle checks cover finite attributes, winding, labels and bounds; they do not certify architectural codes or physical dynamics.

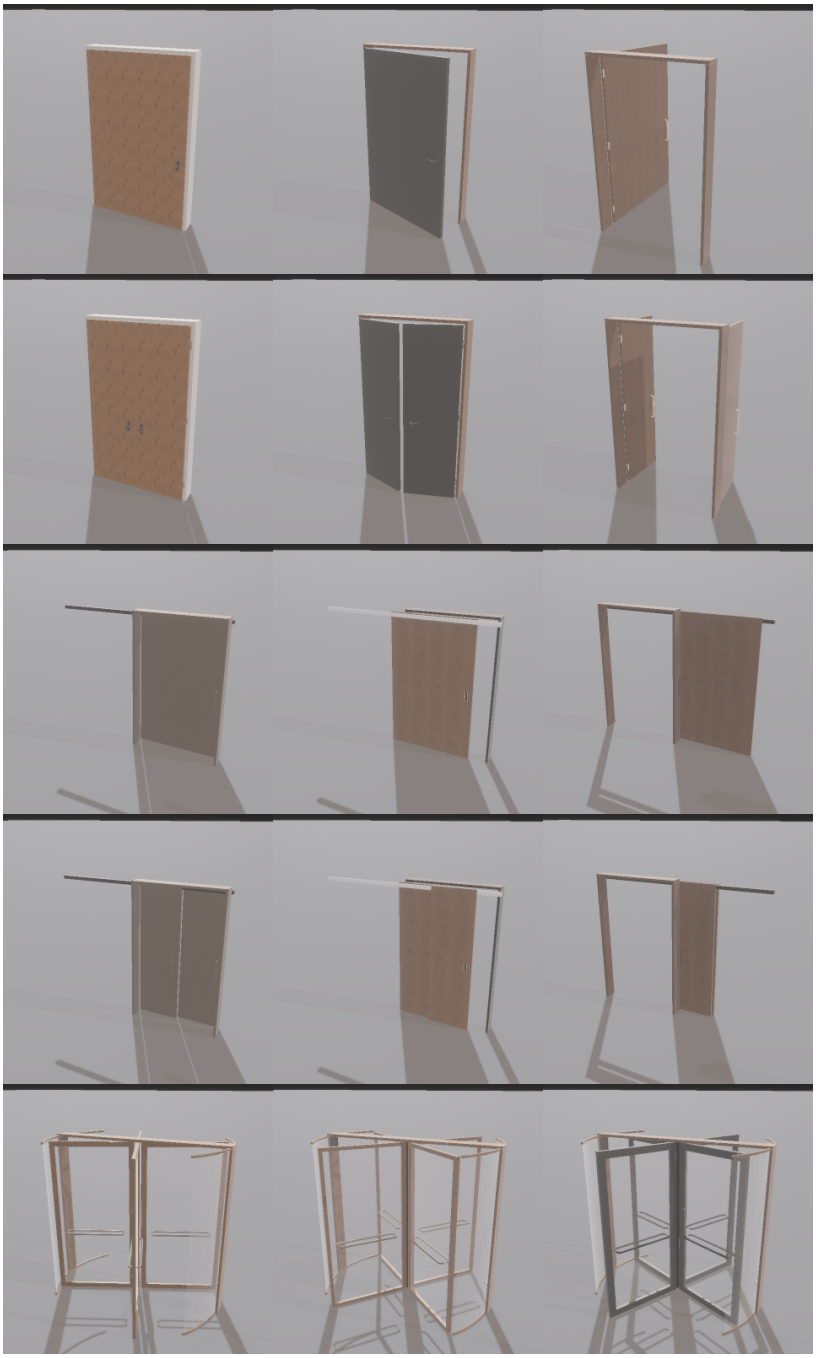


Figure 24: Architectural door construction. Rows: hinged, paired, sliding, telescoping and revolving. The first four rows show sampled shut, ajar and open states; revolving rows show static rotor phases. Panel and finish parameters also vary.



Figure 25: Cabinetry construction. Rows: office storage, counter base, base-and-overhead cupboard, pantry and sink base. Columns are prespecified seeds 0, 3 and 7, without visual filtering. All six examples and semantic/normal channels are available in the project-page category viewer.

8 Procedural material space

8.1 Production bindings and sampling protocol

The material atlas covers all 36 architectural/object surface roles, with the Floor role split into timber, carpet and stone, plus a dedicated rug program: 39 rows. Each row records 256 consecutive material seeds, for 9,984 recipe records. The finish selector is the production seeded selector, not a round-robin assignment. Six fixed illustration seeds (0, 1, 4, 19, 53, 101) supply 234 native PBR renders, matched normal renders and base-color, tangent-normal and roughness map previews. Seed four is included to illustrate frosted interior glass. The full consecutive cohort, rather than these illustrations, defines the reported frequencies; failed seeds are not replaced.

Each render uses the actual `IndoorMaterials` factory and `for_part` binding, including palette tint, texture structure group, UV transform, clearcoat, anisotropy and transmission. The studio reuses a native Bevy renderer. Texture maps are baked by the production CPU reference implementation; this study does not measure GPU-baking throughput. The rig uses fixed 1,400/250 lux directional lights with shadows, the same seed-zero procedural reflection environment, indoor capture exposure and a 40° vertical field of view. A 0.46 m diameter sphere and a 0.46×0.49 m beveled panel expose curved highlights and flat detail. Leaves use production blades with normalized atlas UVs. The backdrop is identical across columns and reveals glass transmission. These are material probes, not representative furnished rooms.

The separate assignment audit constructs every primary-room object in 256 consecutive mixed-layout rooms at density 0.65, without humans: 15015 objects and 53927 material parts. Rug fields use four bounded dye/pile slots; their bindings retain upholstery. Generic object finishes must have a valid slot in $[0, 12)$ and resolve from their recorded finish seed; deliberately independent book-cover slots are retained. Food uses separately validated role-specific bindings. Per-object-category surface and slot counts are exported. This cohort checks application frequency; the studio does not estimate it. Architectural walls/floors, human skin/hair, and installation-specific facade privacy treatments are outside the object-assignment denominator. Human complexion/groom variation is not represented by these substrate probes.

8.2 Correlated controls, bounded palettes

Within a room, twelve finish slots share at most three texture structures per surface. Ceramic, Fabric, FabricAlt and Leather have three structures; Wood, WoodEdge, Leaf, LeafLight, LeafVariegated and Succulent have two. Other roles reuse one structure; rug fields have up to four dedicated dye/pile structures. Shared map handles bound synthesis/upload costs; independent room material seeds resample the underlying recipe. These are correlated palettes, not twelve independent textures. Painted Metal is dielectric; Chrome is conductive. Screen, phone, television, artwork, printed-paper and whiteboard maps come from their dedicated content programs. Container glass and liquids retain several fixed optical constants; their presence is not evidence of broad optical diversity.

The page pairs each image with its actual PBR scalars and seed, and exposes eight-bin observed parameter marginals and object-assignment histograms. Optional subprograms keep their own denominators. Recorded ranges include conditional controls and are not causal influence measurements. Raw base-color maps precede the material multiplier; roughness previews show the packed green channel before its scalar multiplier (or the scalar when no texture exists). Tangent normals are distinct from rendered view-space normals. The maps include production mip chains, while map previews show level zero. A fixed RGB image or parameter count cannot establish perceptual volume, photographic realism, or 10M-sample learning utility. No product-of-ranges diversity estimate is reported.

8.3 Role inventory and realized object use

Table 13: Current material roles. Structures are the maximum shared map programs per room. Object parts and occupied finish slots are observed in the 256-room assignment cohort; zero means absent from that object-only cohort, not an unsupported material.

Surface / floor type	Structures	Object parts	Slots seen
Paint	1	100	0
Accent	1	0	0
Wood	2	2495	12
WoodEdge	2	3073	12
FloorTimber	1	0	0
FloorCarpet	1	0	0
FloorStone	1	0	0

Surface / floor type	Structures	Object parts	Slots seen
Ceiling	1	0	0
Metal	3	9159	12
Chrome	3	2189	12
Plastic	3	8307	12
Fabric	3	590	12
FabricAlt	3	1724	12
Glass	1	0	0
Ceramic	3	1456	15
Soil	1	1116	0
Leaf	2	736	12
LeafLight	2	721	12
Paper	1	4212	13
Screen	1	426	12
Ink	1	5682	0
Light	1	451	0
Concrete	1	413	0
Art	1	1783	12
Rubber	1	5133	0
LeafVariegated	2	72	12
Terracotta	1	333	1
Bark	1	736	0
GlassInterior	1	57	0
ContainerGlass	1	399	0
Liquid	1	135	0
Drink	1	406	0
PhoneScreen	1	368	12
PrintedPaper	1	304	12
Leather	3	437	12
Whiteboard	1	232	12
Television	1	168	12
Succulent	2	308	12
Rug	4	206	4

Reserved fruit, crust and bread-scoring finishes are counted alongside generic object finishes, using a separate binding namespace. They retain role-specific pigments rather than arbitrary ceramic or paper colors.

8.4 Control families

Paint, accent and ceiling Correlated scene pigment; roller, stipple, knockdown and trowel application; deposit scale, coverage, pinholes, gloss and local roughness.

Timber and bark Growth-ring warping, longitudinal/transverse grain, cut width, knots, stain pigmentation, finish absorption, pores and coating. Floor planks additionally vary board/joint scale.

Textiles and leather Weave/knit structure, yarn count and crimp, dye/pattern, pile and lustre; leather cell scale/stretch, grain, wrinkles, wear and coating. Carpet/rug pile and dye fields use separate room-scale programs, while upholstery resolves individual yarn crossings. Floor installations vary bond, basket strips, cut tint and roughness.

Concrete, stone, soil and terracotta Aggregate/grain distribution, deposits, veins, inclusions, polish and relief. The FloorStone row samples mineral mixtures rather than treating every tile as marble.

Ceramic Body/glaze mixture, pigment, gloss, clearcoat, firing variation, speckles, pooling, pinholes and crackle. Tint and texture structure are separately bound by the production factory.

Botanical surfaces Blade-aligned venation, chlorophyll patches, variegation, wax and restrained diffuse transmission. Succulent cuticles use a separate tileable pore/bloom program without blade venation.

Metal, plastics, glass and content Conductive versus painted bases, roughness and directional reflection; glass index, tint/attenuation, thickness and clear/frosted optics; independently seeded procedural screen/print content. Some constants are intentionally fixed by role.



Figure 26: Native PBR material probes: Paint, Accent, Wood, WoodEdge, FloorTimber, FloorCarpet. Columns share seed, camera, lights and geometry within each row; correlated production recipes and finish slots vary. Null slots use the base role. Six fixed examples illustrate, rather than estimate, the sampled distribution. Inspect the complete 256-seed marginals, map channels and actual PBR multipliers in the project-page atlas.

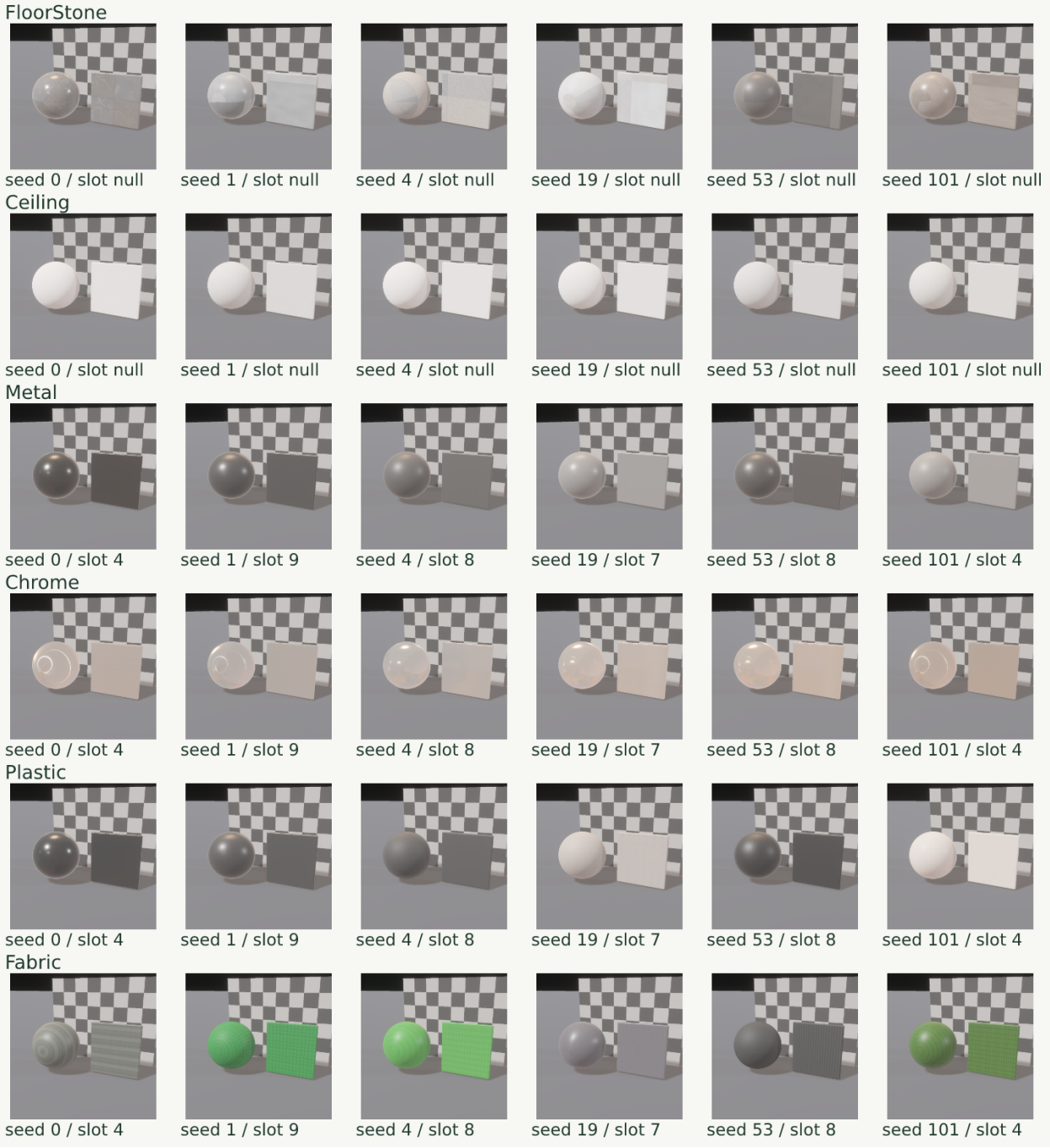


Figure 27: Native PBR material probes: FloorStone, Ceiling, Metal, Chrome, Plastic, Fabric. Columns share seed, camera, lights and geometry within each row; correlated production recipes and finish slots vary. Null slots use the base role. Six fixed examples illustrate, rather than estimate, the sampled distribution. Inspect the complete 256-seed marginals, map channels and actual PBR multipliers in the project-page atlas.

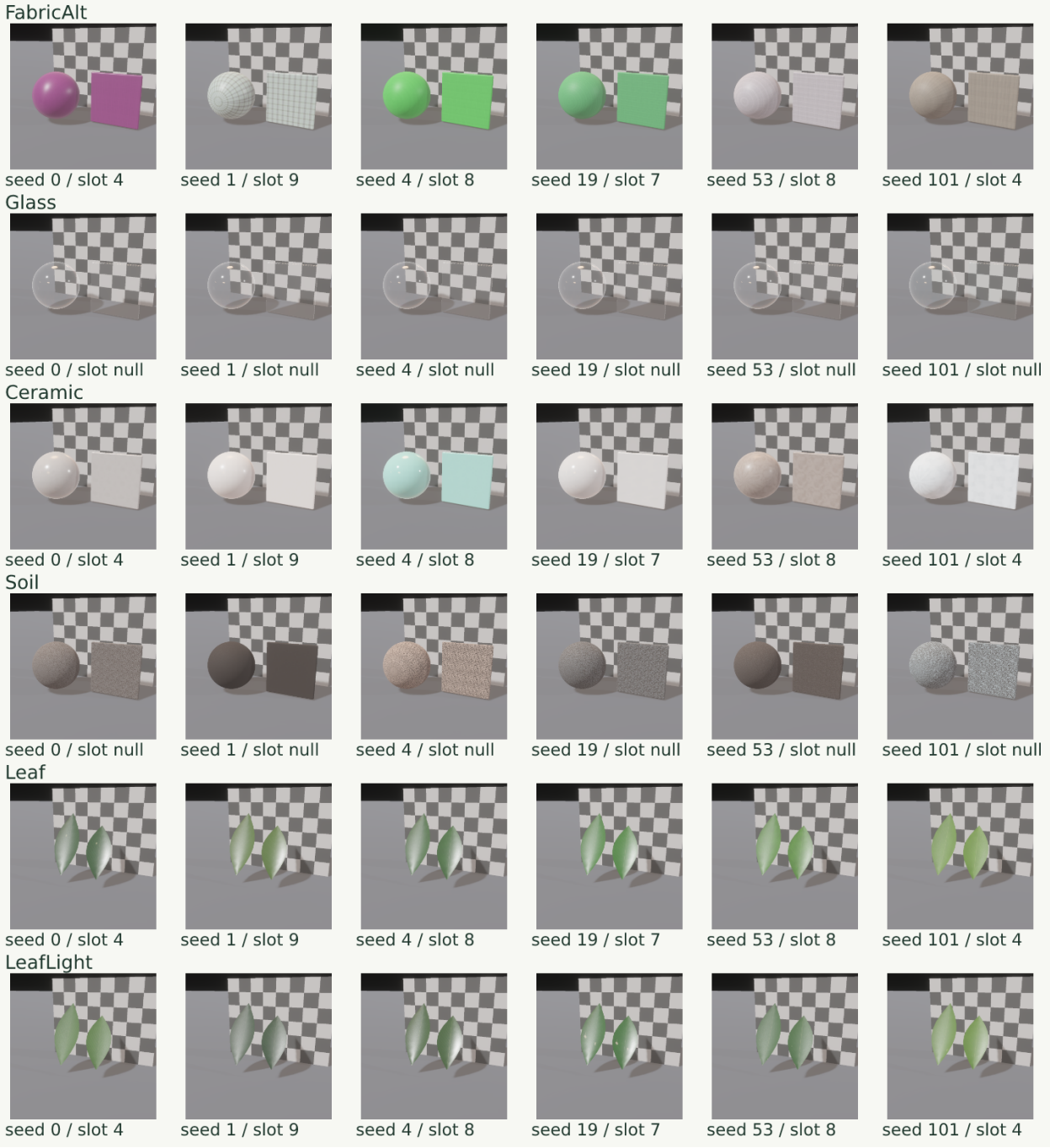


Figure 28: Native PBR material probes: FabricAlt, Glass, Ceramic, Soil, Leaf, LeafLight. Columns share seed, camera, lights and geometry within each row; correlated production recipes and finish slots vary. Null slots use the base role. Six fixed examples illustrate, rather than estimate, the sampled distribution. Inspect the complete 256-seed marginals, map channels and actual PBR multipliers in the project-page atlas.

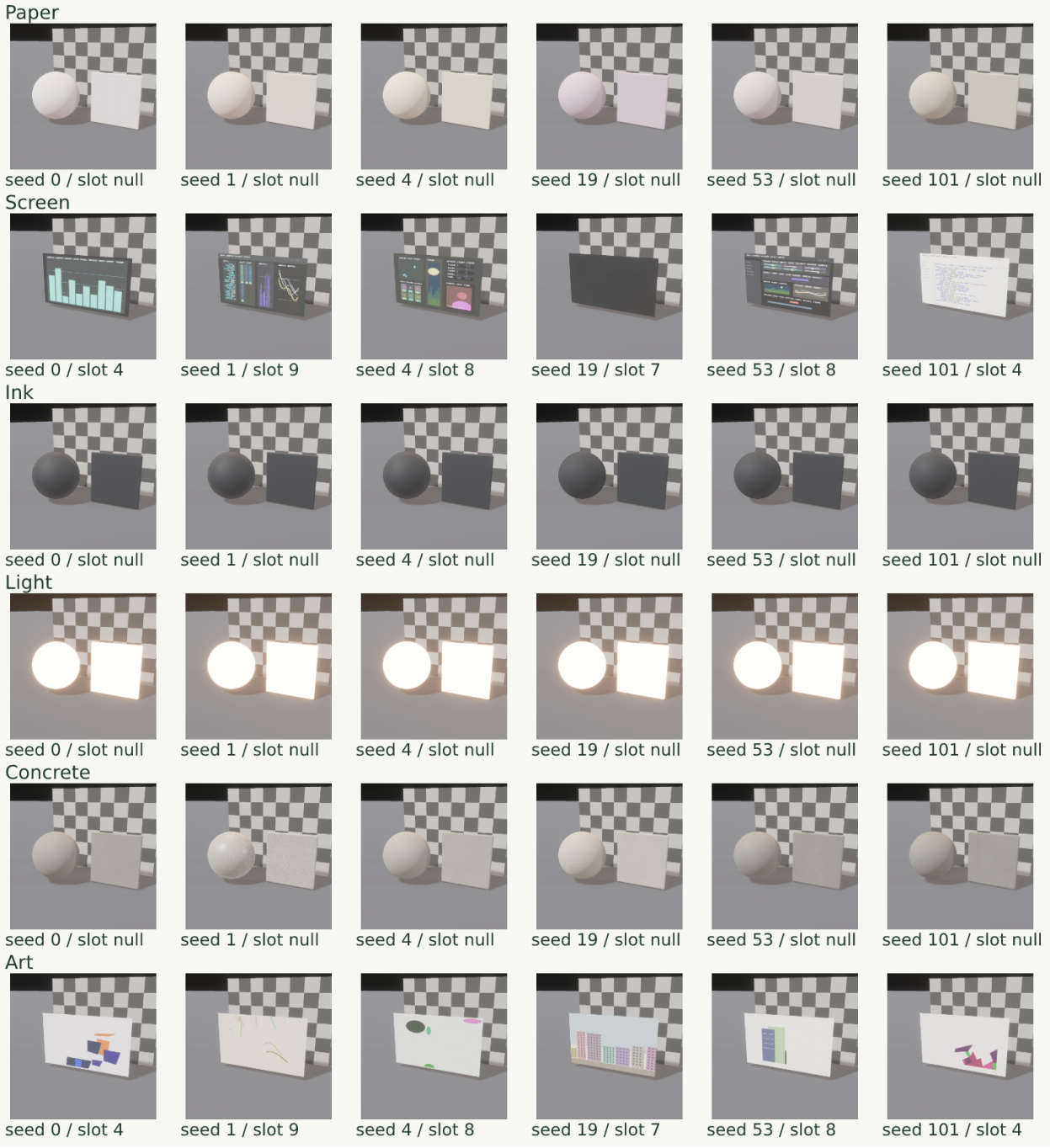


Figure 29: Native PBR material probes: Paper, Screen, Ink, Light, Concrete, Art. Columns share seed, camera, lights and geometry within each row; correlated production recipes and finish slots vary. Null slots use the base role. Six fixed examples illustrate, rather than estimate, the sampled distribution. Inspect the complete 256-seed marginals, map channels and actual PBR multipliers in the project-page atlas.

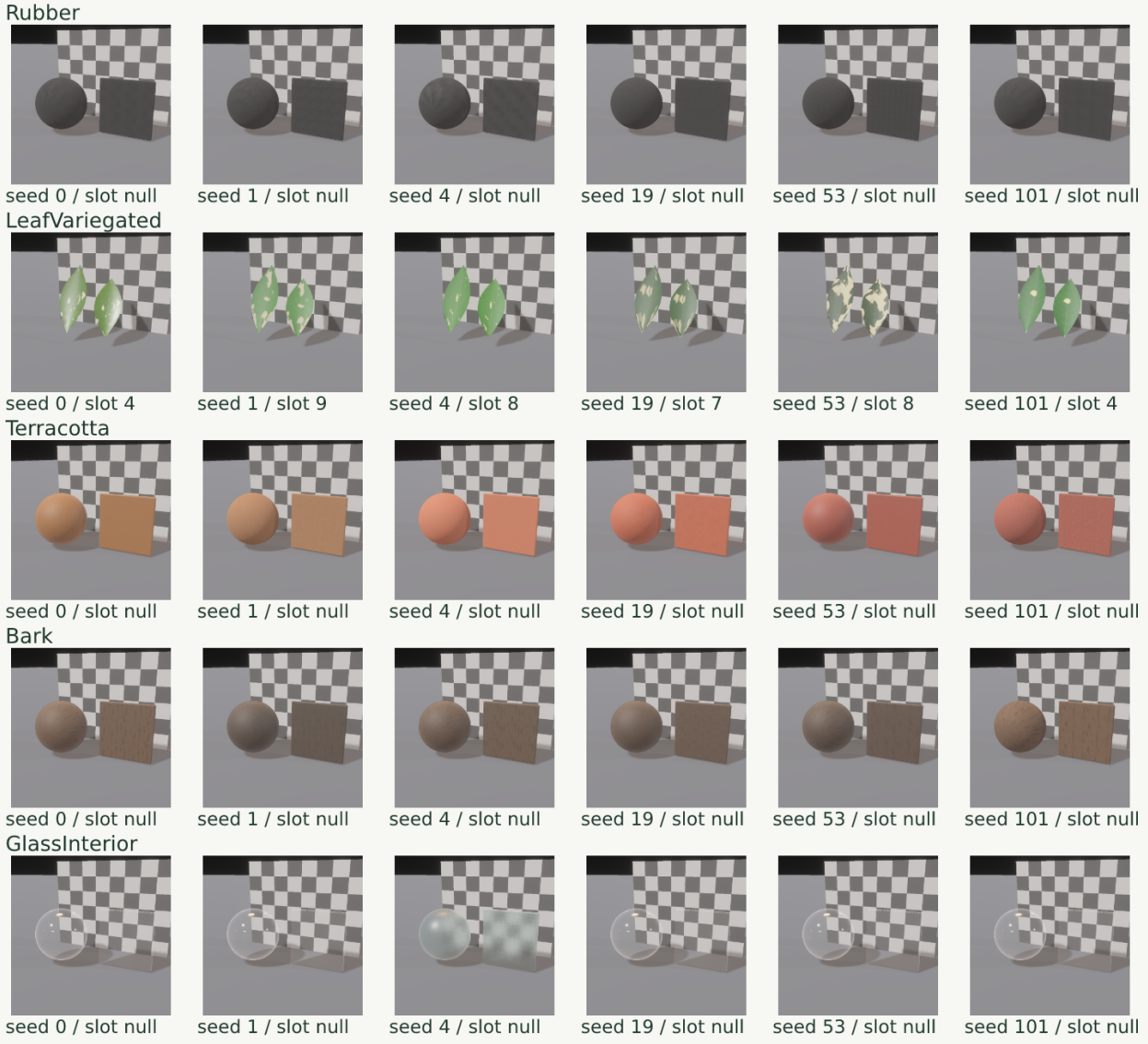


Figure 30: Native PBR material probes: Rubber, LeafVariegated, Terracotta, Bark, GlassInterior. Columns share seed, camera, lights and geometry within each row; correlated production recipes and finish slots vary. Null slots use the base role. Six fixed examples illustrate, rather than estimate, the sampled distribution. Inspect the complete 256-seed marginals, map channels and actual PBR multipliers in the project-page atlas.

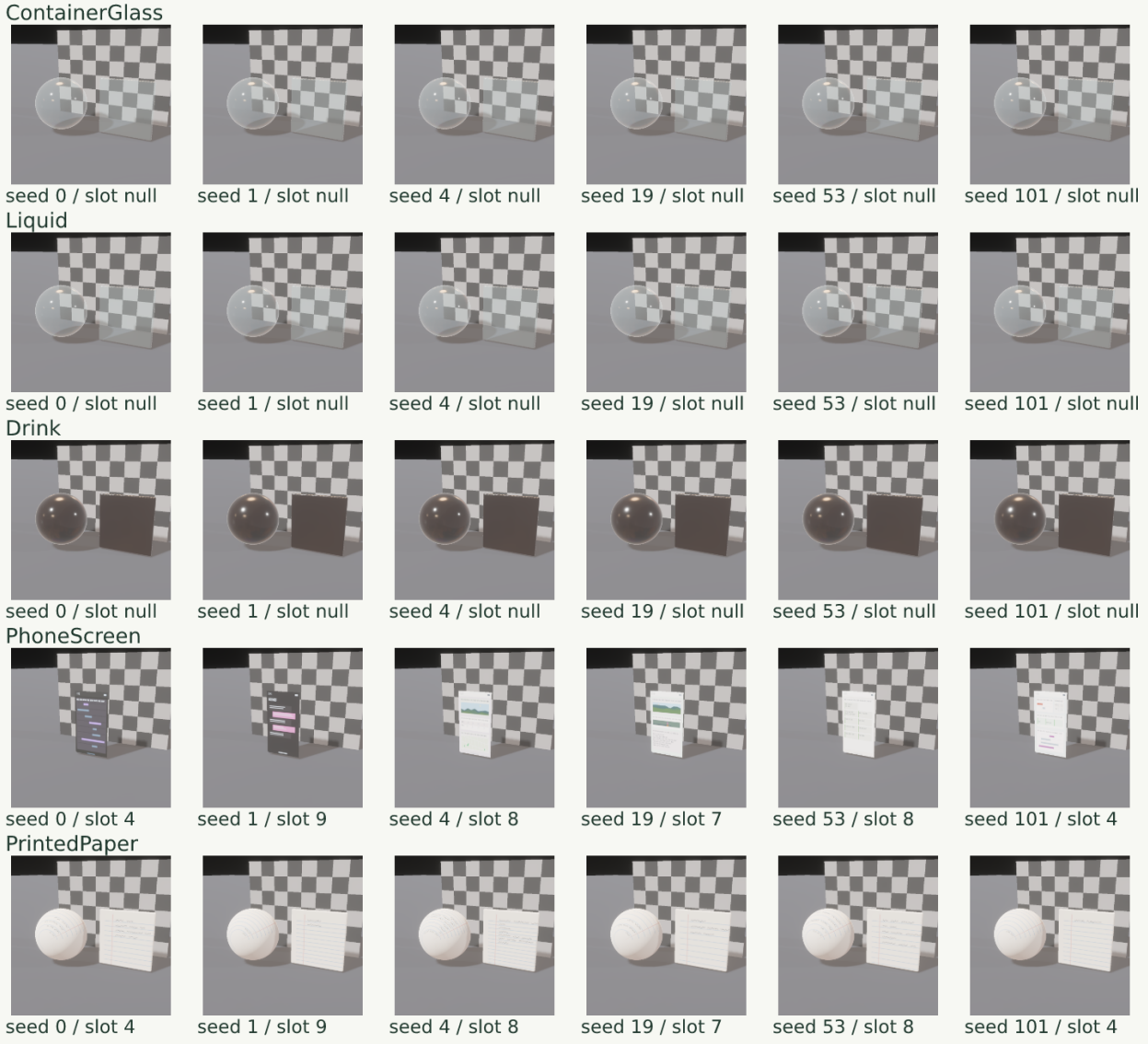


Figure 31: Native PBR material probes: ContainerGlass, Liquid, Drink, PhoneScreen, PrintedPaper. Columns share seed, camera, lights and geometry within each row; correlated production recipes and finish slots vary. Null slots use the base role. Six fixed examples illustrate, rather than estimate, the sampled distribution. Inspect the complete 256-seed marginals, map channels and actual PBR multipliers in the project-page atlas.

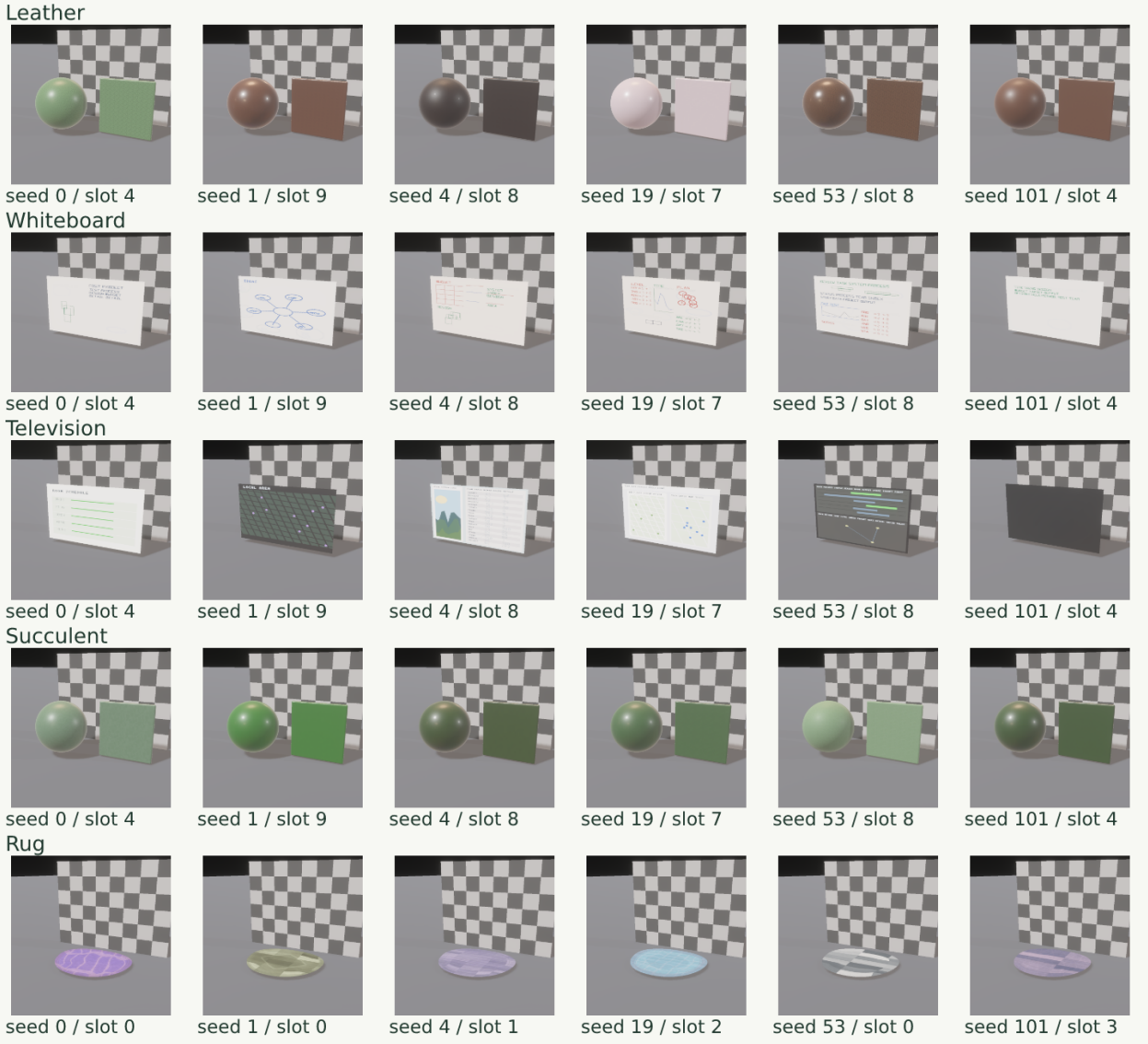


Figure 32: Native PBR material probes: Leather, Whiteboard, Television, Succulent, Rug. Columns share seed, camera, lights and geometry within each row; correlated production recipes and finish slots vary. Null slots use the base role. Six fixed examples illustrate, rather than estimate, the sampled distribution. Inspect the complete 256-seed marginals, map channels and actual PBR multipliers in the project-page atlas.

8.5 A parameter-linked example: ceramic

The following columns are the same six Ceramic images in the material plates. Recipe controls and final shader multipliers are shown separately. Glaze gloss drives the coating program; the roughness map retains local spatial detail. This is a correlated draw, not six isolated gloss interventions.

Material seed	0	1	4	19	53	101
Texture structure	1.000	0.000	2.000	1.000	2.000	1.000
Glaze gloss	0.792	0.624	0.948	0.840	0.412	0.184
Crackle strength	0.000	0.000	0.135	0.230	0.000	0.000
Relief (mm)	0.059	0.024	0.052	0.059	0.020	0.017
Shader roughness multiplier	0.943	0.867	0.980	0.927	0.907	0.933
Shader clearcoat	0.206	0.256	0.419	0.257	0.179	0.051

Table 14: Recorded recipe values and actual PBR bindings for the Ceramic grid. Full seed records and observed ranges for every material category accompany the project page.

9 Procedural capacity and sampling scale

The number of seed values, the number of named object categories, and the number of useful training examples are different quantities. We define a configuration-conditioned capacity framework and exact analytical reference cases. We do not estimate the generator's total perceptual support or a downstream learning exponent from the finite cohorts in this paper.

9.1 Population, unit and distinguishability

Fix resolved settings C, engine and asset identity E, and seed law π. Generation $G_E(C, s)$ includes its bounded search and recovery policy. Report success yield $a(C)$ separately from the distribution of accepted outputs; failed rooms are not useful diversity classes. An observation descriptor is $X = \Phi(\mathcal{R}_E(G_E(C, s)))$, conditional on success, where $\mathcal{R}$ includes the declared views, times, resolution, exposure and annotation policy. A frozen partition Q_ϵ assigns observations to distinguishable classes with probabilities $p_j = \Pr[Q_\epsilon(X) = j \mid \text{success}]$. The representation and resolution are part of the result, not hidden defaults.

A structural room family, dressed scene, multi-view capture and temporal clip are different sample units. Camera, material and time variants can add useful observations without adding independent structural families. Confidence intervals and train/validation splits must retain this grouping. A single deterministic 64-bit root seed gives an input-support ceiling of 2^{64} for fixed settings; domain-separated streams do not increase its entropy. Extra independent stochastic keys count only if the population law declares them. Ideal continuous ranges do not imply infinitely many distinguishable digital images. Finite precision, conditional branches, clearance rejection, occlusion and pixel resolution reduce realized variation.

The resolved population must include every active factor: architecture and functional program; furniture, props and accepted support/interaction relations; human bodies, clothing, hair and posed joints; assigned material/texture programs; illumination and exposure; and captured camera intrinsics, extrinsics and times. Compute joint descriptors on the same accepted rooms. Multiplying marginal category counts or entropies assumes independence that the generator does not have. Hidden material recipes, rejected objects, seed IDs and unsampled motion endpoints contribute no observed diversity. Report conditional factor coverage alongside the aggregate so broad lighting variation cannot conceal repeated furnishings or people.

The current Mixed sampler (also accepted as `all`) proposes each of the 20 concrete room programs with probability $1/20$, including kitchens, dining, teaching, transit, exhibition and public suites. A suite is one room draw even when it contains several functional zones. Measure accepted proportions and failure yield separately. Dataset defaults conditional on `procedural-indoor` capture four 256×256 RGB views at progress zero. Camera trajectories are enabled, but one captured timestep supplies no temporal variation; ARDY and object motion are disabled unless requested. A longer path does not multiply sample count. The three-view 512×512 library convenience preset and the paper's recorded four-view, two-time gallery are different populations and require distinct capacity receipts.

Geometry partitions should use primary-region topology and actual surface or support descriptors, excluding random IDs and seeds. Appearance and learned image descriptors need their own definitions. Pairwise distance below a threshold is generally not an equivalence relation: connected-component chaining can join visually different endpoints.

Occupancy analysis therefore requires an explicit partition, frozen independently of the final audit. Direct near-pair rates remain separate diagnostics. Codebook identity, metric units, canonicalization and out-of-domain behavior must accompany every estimate.

9.2 Effective numbers and exact occupancy identities

Following the effective-number interpretation of diversity [4], define

$$D_0 = |\{j : p_j > 0\}|, \qquad D_1 = \exp\left(-\sum_j p_j \log p_j\right), \qquad D_2 = \left(\sum_j p_j^2\right)^{-1}. \tag{1}$$

These respectively describe support, entropy-equivalent balance and collision-equivalent balance; $D_2 \le D_1 \le D_0$. They coincide for a uniform finite population. A large rare tail can increase D_0 with little effect on D_2. Multiplying marginal parameter counts or entropies ignores dependence. For a correctly factorized program graph, joint entropy is $H(Z) = \sum_k H(Z_k \mid \mathrm{parents}_k)$, and a deterministic rendering and partition cannot increase entropy beyond the stochastic inputs.

For $n \ge 1$ independent successful draws from the same population, let $U(n)$ be expected occupied classes and $c_2 = \sum_j p_j^2 = 1/D_2$. Indicator expectations give

$$U(n) = \sum_j \left[1 - (1 - p_j)^n\right], \tag{2}$$

$$U(n+1) - U(n) = \sum_j p_j (1 - p_j)^n, \tag{3}$$

$$\mathbb{E}[\text{equal-class pairs}] = \binom{n}{2} c_2. \tag{4}$$

There is also a distribution-independent bound, for this fixed partition and IID law,

$$\frac{n}{1 + (n-1)c_2} \;\le\; U(n) \;\le\; \min(n, D_0). \tag{5}$$

To see the lower bound, let N_j be sample class counts and K_n their number of nonzero entries. Cauchy–Schwarz gives $K_n \ge n^2 / \sum_j N_j^2$. Jensen's inequality and $\mathbb{E}\sum_j N_j^2 = n + n(n-1)c_2$ give Equation 5. It is a bound, not an occupancy model inferred from D_2 alone. The two known populations in Figure 33 have equal D_2 but different support and novelty curves.

With uniform support S, Equation 2 becomes

$$U(n) = S\left[1 - (1 - 1/S)^n\right] \simeq S\left(1 - e^{-n/S}\right). \tag{6}$$

The first repeated class and substantial saturation are different scales: expected colliding pairs reach one near $\sqrt{2D_2}$, whereas uniform occupancy saturates on scale S. Ten million draws with roughly 95% distinct classes would require about one hundred million equiprobable classes under this particular model. Neither uniformity nor that support is established for Zeroverse. The analytical table is generated in Rust from numerically stable logarithm/exponential formulas, not fitted to captures.

Known uniform support	Expected distinct	Distinct/draws	Next-draw novelty
1 million	1.000 million	10.00%	0.00%
10 million	6.321 million	63.21%	36.79%
100 million	9.516 million	95.16%	90.48%

Table 15: Exact uniform-occupancy formulas at $n = 10^7$, rounded for display. No row estimates the support of this generator.

Sampling without replacement requires a different law. For a finite seed pool of size M with class multiplicities m_j, replace $(1-p_j)^n$ by $\binom{M-m_j}{n}/\binom{M}{n}$. Consecutive-seed regression audits are useful deterministic coverage checks but do not by themselves justify IID confidence claims. Stratified and family-grouped samples require their declared weights and uncertainty method. Capture failures and retries must not silently change the reported target population.

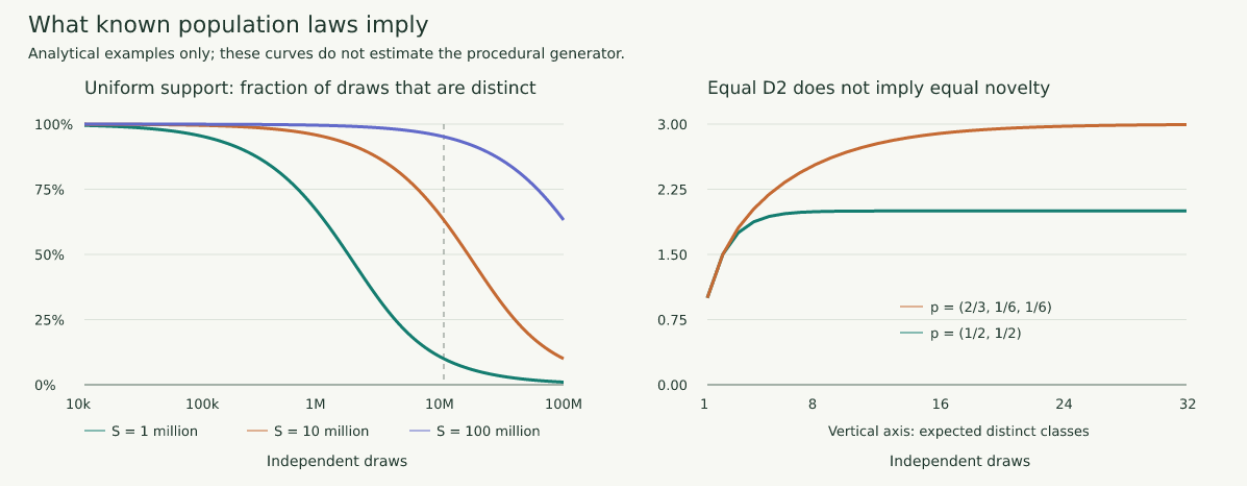


Figure 33: Analytical occupancy examples, not measured Zeroverse capacity. Left: distinct fraction for uniform populations with known support; dashed line marks ten million draws. Right: two populations have the same collision diversity $D_2 = 2$ but different occupancy curves and limiting support. A single effective number does not determine long-run novelty.

9.3 Estimation and insufficient evidence

Given IID class counts n_j in $n \geq 2$ observations,

$$\widehat{c}_2 = \frac{\sum_j n_j(n_j - 1)}{n(n-1)} \tag{7}$$

is unbiased for c_2; $1/\widehat{c}_2$ is not unbiased for D_2. Pairs that share an observation are dependent. Treating all $\binom{n}{2}$ pair indicators as independent binomial trials gives unjustified confidence bounds. Zero observed matches must not produce a claim of infinite capacity.

A conservative fallback uses $m = \lfloor n/2 \rfloor$ prespecified disjoint pairs of independently sampled rooms. If none match, the exact one-sided $1 - \alpha$ upper confidence bound on c_2 and corresponding lower bound on D_2 are

$$c_{2,\text{upper}} = 1 - \alpha^{1/m}, \qquad D_{2,\text{lower}} = \frac{1}{1 - \alpha^{1/m}}. \tag{8}$$

For 512 rooms and $\alpha = 0.05$, this fallback yields only about 86 as a lower bound on collision diversity, even when every observed class is distinct. It is deliberately conservative, not an optimal use of all pairs. A stronger dependency-aware estimator or a uniform birthday model needs its own justification. No finite support upper bound follows from this observation without further population assumptions, apart from any separately declared finite input ceiling.

Report observed support, singleton/doubleton counts, sample coverage, rarefaction and held-out next-block novelty. Unseen-species theory permits useful bounded extrapolation, but also establishes limits on distribution-free prediction horizons [5]. It does not justify projecting hundreds of rooms to ten million by fitting a convenient curve. Quantizers and thresholds must be frozen before final evaluation; resampling and uncertainty operate on independent room families. Ordinary bootstrap resampling of an all-unique observed set cannot discover an unobserved rare tail. Adaptive stopping and multiple reported configurations/resolutions also need an explicit confidence policy.

9.4 Similarity, resolution scaling and learning

For continuous image features, an accompanying sample diagnostic can use a positive semidefinite similarity kernel with $K_{ii} = 1$, such as $K_{ij} = \exp(-\|x_i - x_j\|^2/(2\sigma^2))$ on frozen SigLIP2 features. If λ_i are eigenvalues of K/n, the Vendi score is $\exp(-\sum_i \lambda_i \log \lambda_i)$; its order-two analogue is $1/\sum_i \lambda_i^2$ [6, 7]. These measure similarity diversity of the observed sample, depend on bandwidth and sampling, and are at most n. They are not total generator support or the number of statistically independent training examples. A linear feature kernel is additionally rank limited by embedding dimension. Centered covariance rank and ellipsoid-volume diagnostics have different meanings and must not be substituted for class counts.

At declared structural resolutions, a candidate local relation is $\log D_2(\epsilon) \approx a - d \log \epsilon$. This is a resolution-scaling hypothesis over a tested interval, not a global dimensionality theorem or a sample-count law. Conditional branches and finite precision can produce plateaus and changing slopes. Evaluate several frozen resolutions and held-out cohorts before reporting an exponent. A separate downstream hypothesis,

$$L(n) = L_\infty + A\,U(n)^{-\beta}, \tag{9}$$

would require fixed model, optimizer, compute, real validation set and repeated learning experiments. Compare it against raw-count and family-count alternatives; SigLIP spread cannot identify β or prove transfer. No fitted coefficients for either scaling hypothesis are claimed here.

9.5 Qualification and API contract

The implemented capacity query binds resolved configuration, seed/failure law, sample unit, exact engine/model identities, partition/kernel and resolution, confidence policy, and requested sample horizon. It returns observed statistics, bounds or explicitly model-conditional forecasts, with an optional point estimate and nullable unresolved bounds. No matching evidence returns *not qualified*, not a guessed dataset size. A read-only query must not initialize models or render scenes; qualification is a separate explicit operation. These configuration-bound API types live in the portable capture crate, the generator supplies the typed configuration adapter, and the publication crate performs explicit offline qualification and generates receipt-bound tables, figures and HTML. The initial estimator supplies joint fixed-bin counts, observed-cohort rarefaction and exact Clopper–Pearson collision bounds using disjoint room pairs. It uses a consistent two-sided tail allocation, adjusted across prespecified definitions; choosing a one-sided interval only after seeing zero matches would under-cover near that selection boundary. Equation 8 remains the separately prespecified one-sided analytical example. No unseen-tail forecast is promoted. Native capture observation verifies full calibration, poses, frame delivery and finite RGB/geometry/semantic buffers; CPU construction alone cannot qualify rendered appearance. Bounded persistent workers preserve draw order and failed attempts, and ordinary rendering pays no capacity-audit cost. The analytical illustration command remains distinct from empirical qualification.

The optional image-space audit freezes the capture index, verified SigLIP2 weights, preprocessing and RBF bandwidth before inference. It averages normalized view embeddings within each room and normalizes the resulting centroid, giving every room equal weight. The exact bounded eigensolver checks its residual and the positive-semidefinite, trace-one spectrum of K/n; its Vendi and participation scores describe the observed sample only. Complete fixed-seed replay is tested separately from different-seed novelty. Whole-image embeddings include exterior context and do not substitute for the primary-region geometry observer.

The validation sequence is: exact finite-population oracles; independent pilot calibration of geometry/appearance equivalence; randomized structural audits; nested rendered cohorts; held-out novelty forecasts; then fixed-budget downstream learning. All views and times of a family stay together. Qualify whole-image and primary-region measurements separately so background changes cannot stand in for useful room diversity. Increase sample budgets only when interval precision or unexplored strata require them, and expose the largest tested forecast horizon. The Rust analytical fixtures enumerate small probability trees, verify occupancy and collision identities, check rare-tail bounds and zero-match confidence arithmetic, and regenerate the figure/table deterministically. They ground the formulas; they do not experimentally qualify capacity of the rendered generator.

10 Limitations

Procedural correctness, broad parameter variation and efficient capture do not establish photographic realism. Hair is a surface groom, skin lacks a full subsurface model, some furnishings remain sparse, object motion uses approximate collision proxies, and diffuse GI uses probes and simplified transport. Glass and lighting are approximations rather than full path tracing. Transparent panes are excluded from opaque shadow and diffuse-transport geometry, so direct sunlight does not acquire physically complete tint, absorption or frosting redistribution. Geometric through-glass labels deliberately use straight rays rather than refracted RGB correspondence. Finite path-traced reference diagnostics still expose substantial scene-dependent transport error; the camera-baseline sweep does not measure physical accuracy or visual plausibility.

The current study is a generator audit and finite capture cohorts, not a ten-million-scene run. Process-lifetime memory stability, including retained graphics-driver allocations, is not qualified by these finite cohorts. The evaluation does not establish balanced coverage of all rare architectural combinations, collision-free arbitrary motion, suitability for every reconstruction loss, or transfer to real photographs. Shared-view defaults deliberately correlate cameras within a room; train/validation splits should therefore be made by room seed, not by image. A downstream assessment should freeze a training budget and real evaluation set, compare matched data quantities and camera policies, and measure

reconstruction/calibration quality alongside redundancy. The manifests and metrics make that experiment reproducible without claiming its outcome in advance.

References

[1] D. Xie et al. LRM-Zero: Training Large Reconstruction Models with Synthesized Data. NeurIPS, 2024. https://arxiv.org/abs/2406.09371.

[2] Y. Chen et al. MVSplat: Efficient 3D Gaussian Splatting from Sparse Multi-View Images. ECCV, 2024. https://arxiv.org/abs/2403.14627.

[3] M. Tschannen et al. SigLIP 2: Multilingual Vision-Language Encoders with Improved Semantic Understanding, Localization, and Dense Features. 2025. https://arxiv.org/abs/2502.14786.

[4] M. O. Hill. Diversity and Evenness: A Unifying Notation and Its Consequences. Ecology, 54(2):427–432, 1973. https://doi.org/10.2307/1934352.

[5] A. Orlitsky, A. T. Suresh and Y. Wu. Estimating the number of unseen species: A bird in the hand is worth $\log n$ in the bush. 2016. https://arxiv.org/abs/1511.07428.

[6] D. Friedman and A. B. Dieng. The Vendi Score: A Diversity Evaluation Metric for Machine Learning. TMLR, 2023. https://arxiv.org/abs/2210.02410.

[7] A. P. Pasarkar and A. B. Dieng. Cousins of the Vendi Score: A Family of Similarity-Based Diversity Metrics for Science and Machine Learning. AISTATS, PMLR 238:3808–3816, 2024. https://proceedings.mlr.press/v238/pasarkar24a.html.

bevy_zeroverse: calibrated fisheye cameras

Mitchell Mosure

October 2026

1 Projection and sampling

The optional central angular camera uses $r(\theta) = \theta(1 + k_1\theta^2 + k_2\theta^4 + k_3\theta^6 + k_4\theta^8)$, with $\theta = \text{atan2}(\sqrt{x^2 + y^2}, -z)$. Full pixel-space K, image dimensions, half-pixel centers, model version, coefficients and calibrated angular domain are exported per view. Monotonicity is certified over the entire domain using interval derivative bounds. The coefficient order matches OpenCV fisheye (Kannala–Brandt), verified against its forward-hemisphere equations; OpenCV integer-centered pixels require subtracting 0.5 from the exported principal point and converting the view basis to right/down/forward. The inverse supports rays behind the optical plane, beyond the domain of OpenCV routines based on the front-hemisphere arctangent. This is a central camera model, not rolling shutter or a non-central lens.

Lens assignment is seeded by scene and camera index before placement retries. Configuration controls its Bernoulli probability, continuous diameter FOV, polynomial variation, anisotropy, principal-point displacement and circular, full-frame/diagonal or explicitly elliptical fit. Circular fits enforce equal focal axes and a contained disk; elliptical fits contain the sampled ellipse; diagonal fits cover every sensor corner, including principal-point shifts. Defaults remain pinhole; enabling the lens program defaults to probability 0.25, a 140–210 degree diameter range and 0.2 circular probability. In 1,024 consecutive seeds with 12 slots each, 3062 of 12288 assignments are angular and 625 of those are circular. Calibration validation and exact replay pass for every assignment. The admitted pose distribution is affected by obstacle and shared-surface constraints; lens assignment is preserved. Camera motion and baseline controls remain independent.

2 Rendering and annotation contract

Up to five co-located perspective faces retain the PBR renderer, shadows, lighting and exposure. Guard bands and a GPU angular resolve produce the final image; faces are implementation details and consume no dataset camera indices. SSAA averages RGB in linear light. Final-pixel-center rays independently intersect authoritative annotation triangles in a shared BVH. Fisheye depth is positive ray range; pinhole depth remains positive axial depth. Invalid lens pixels and surface misses are distinct through the calibrated domain mask. Normals are expressed in the logical camera frame; semantic IDs are never interpolated. Glass follows the selected geometric surface policy, not a refracted or reflected RGB light path.

Forward flow traces the preceding capture's surface and projects its corresponding point at the next controlled timestamp. Co-visibility projects each source surface through the target model and tests symmetric tangent-plane agreement at the nearest target pixel. Finite sampling can leave small holes at thin silhouettes; this is the existing dataset visibility policy. Mixed lenses and circular validity use the same projection contract. Only logical cameras contribute bits to the additive membership mask. Burn and Python tensor archives use a versioned mixed-lens contract with per-view model identifiers and five angular parameters; pinhole-only archives preserve version 1. Direct GPU transports retain the same per-view metadata and on-device color conversion.

3 Bounded qualification

The analytic fixture uses a moving occluder, two angular cameras and one pinhole camera over three synchronized captures, with unrelated render updates between captures. Maximum position error is 0.1526 mm; maximum forward-flow error is 0.001817 pixels across 36471 correspondences. There are 48625 co-visibility comparisons

away from silhouette discontinuities and 0 mismatches. These tests exercise occlusion, rearward rays, optical-center displacement and anisotropy.

The furnished-room cohort contains 6 consecutive rooms, three cameras and two timestamps at 512×512, with automatic $2\times$ SSAA and full scene quality. All 36 captures pass self-reprojection (maximum 0.000396 pixels) and unit-normal checks. The realized diameter range is 153.0–209.8 degrees. Figures show actual, uncorrected renderer output; dim and bright scenes remain in the gallery. Geometry and lens settings, calibration, software provenance and artifact checksums accompany every record.



Figure 1: Diagonal and circular fisheye RGB; additive co-visibility from the same circular view (camera 1). Red: camera 0; blue: camera 2; magenta: both; black: no shared peer or invalid lens domain. The full gallery includes all rooms and aligned annotations.

4 Throughput and limits

On an NVIDIA RTX PRO 6000 Blackwell (training paused during capture), the completed-generation rate excluding the first room was 0.517 rooms/s or 3.104 camera/time images/s. The first room took 3.986 seconds and is reported separately. Application startup and PNG/JSON report writing are excluded; rendering, scene work and capture readback are included. These are bounded observations of the recorded workload, not a memory-stability guarantee. Five perspective faces cost more than a single pinhole image; shared scene assets/BVH, indexed-vertex transforms, compact BVH partitioning, nearest-first traversal, stable geometry reuse, and the capture activity gate bound redundant work. The disabled path creates no angular textures or BVH.

Measurements use the optimized development profile (optimization level 3), the default 256 diffuse-GI rays, GPU material synthesis, furnishing density 0.65 and human density 0.25. Desktop applications share the GPU; training is separately identified above. This is native Vulkan qualification. The WebAssembly build passes; browser/Metal runtime and cross-vendor performance remain unqualified for this source.

Screen-space effects remain face-local; the guarded color resolve is not a path tracer. RGB uses perspective-face clipping while exact geometric targets use the calibrated ray range. Geometry within the configured operating range is the qualification domain; grazing near/far boundary behavior, arbitrary optical systems, hardware-vendor parity and downstream learning gains require further qualification. The main architectural cohort retains its original capture provenance; this addendum is independently source-bound.

4.1 Matched camera-model workload

The same six rooms, poses, timestamps, materials, GI, shadows and automatic 2x SSAA are used for both projections. The report rejects differences in the geometry manifest and exported poses. Projection coverage differs; a rectangular pinhole image cannot span the complete fisheye domain. Rates include generation, rendering and CPU readback, with the first room excluded and recorded separately. These are not direct Burn/WGPU or CUDA transport benchmarks.

Workload	Pinhole rooms/s	Fisheye rooms/s
Seven modes, two timestamps	0.815	0.517
RGB, one timestamp	0.979	0.834

4.2 Photometric face-boundary qualification

Identical world rays are compared across all eight adjacent face pairs at fixed exposure, with a continuous environment map. Ambient, shadowed directional, point and spot cases vary the direct illumination while retaining the emissive object and bloom; the combined case also includes SSAO. Glass transmission and metallic reflection are present throughout. Per-boundary and aggregate thresholds are checked. These measurements test raster consistency, not physical accuracy against a path tracer.

Case	Mean error	95th percentile	Maximum
ambient	0.00415	0.02203	0.04535
directional	0.00716	0.03057	0.05395
point	0.00379	0.02073	0.04409
spot	0.00328	0.02055	0.04360
combined	0.00163	0.00645	0.05215

Errors are maximum-channel absolute differences in tone-mapped linear RGB before the composite. Maximum outliers are retained; screen-space and finite shadow-map approximations are not claimed to be exactly invariant across faces. The final composite blends only the guarded overlap. A separate GPU probe executes the production compositor on these rendered textures, approaching each boundary from either side by 10^{-6} radians; every illumination case must have maximum discontinuity below 0.002 and mean below 0.0001. Geometric and categorical targets are not blended.

Remaining photometric limitation. The raw, pre-composite boundary budget did not pass for every pair. This result is retained as a failed diagnostic, even though all final-composite continuity checks passed. View-local postprocessing can produce different radiance estimates for the same ray; blending removes a hard boundary, but does not establish global photometric invariance or path-traced accuracy.